UMC 0.11um Logic and Mixed-Mode Al Advanced Enhancement Vertical PNP BJT SPICE Model (Gummel-Poon) Document

Version 0.4 Phase1
2011/06/01

Table of Contents

List of Figures

List of Tables

1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
t.0_p1	2006/02/27	Y. C. Liu	1. Original Release.
t.1_p1	2009/11/13	Chih-Yang Kao	1. Silicon based model.
0.1_p1	2010/01/08	Jian Zhi Huang	1. Copy model from **l110eal_bjt_vt11.lib** in the G-05-LOGIC/MIXED_MODE11-MMC/AL/L110E-SPICE 2. New DSM version definition.
0.2_p1	2010/07/20	Ho Cheng Hu	1. Version up to 0.2 based on L110AE Si data.
0.3_p1	2010/12/28	Hung Wei Liao	1. Add **4.5x4.5, 9x9, 13.5x13.5** vertical pnp model with SAB
0.4_p1	2011/06/01	Y Z Wang	1. Revised the parameters cje of pnp models without SAB to align vertical pnp models with SAB.

1.2 General Information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

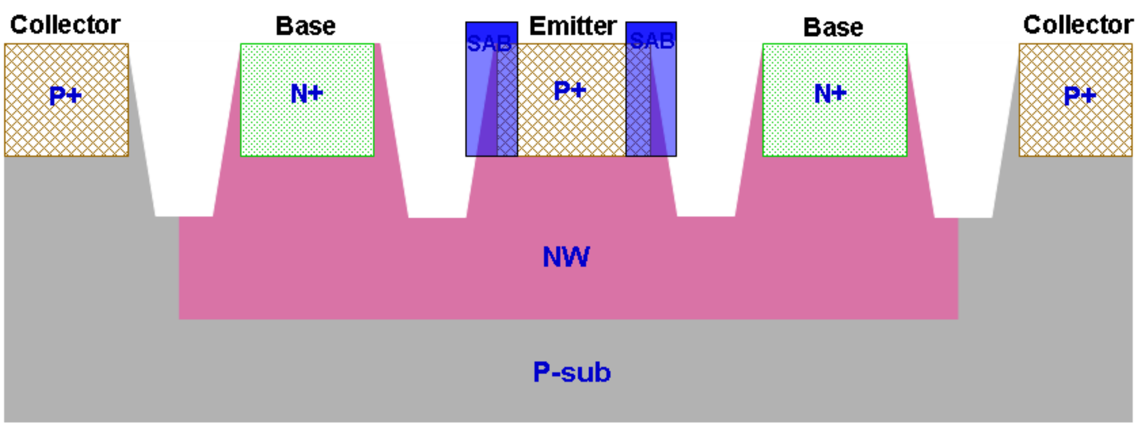

Figure 1-1 BJT Layout of PNP

1.3 Model Usage

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor Graphics Eldo version 6.10_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

TT_BIP:	Typical BJT model
FF_BIP:	Fast BJT model (Maximum Beta model)
SS_BIP:	Slow BJT model (Minimum Beta model)

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE
- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be TT_BIP, FF_BIP, SS_BIP.

For Cadence Spectre
- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator.
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT_BIP, FF_BIP, or SS_BIP.

For Mentor Graphics Eldo
- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName could be TT_BIP, FF_BIP, SS_BIP.

As stated above that BJT model is provided and this model is valid as the following geometry, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry description:

Table 1-2 Vertical BJT device

Vertical PNP	pnp_v45x45_l110e	pnp_v90x90_ l110e	pnp_v135x135_ l110e
Emitter Area	4.5X4.5 μm^2	9X9 μm^2	13.5X13.5 μm^2
Base Area	10.98X10.98 μm^2	15.48X15.48 μm^2	19.98X19.98 μm^2
SAB (Salicide Block) overlap diffusion	No	No	No

Vertical PNP	pnp_sv45x45_ l110e	pnp_sv90x90_ l110e	pnp_sv135x135_ l110e
Emitter Area	4.5X4.5 μm^2	9X9 μm^2	13.5X13.5 μm^2
Base Area	11.7X11.7 μm^2	16.2X16.2 μm^2	20.7X20.7 μm^2
SAB (Salicide Block) overlap diffusion	Yes	Yes	Yes

Temperature range:

-50 °C ~ +125 °C

2 PNP BJT Model Verification

2.1 Silicon Wafer For Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for BJT

	BJT PNP model
Mask ID	BSW0QA-A0
Lot ID	D58YR.1
Wafer Number	03

2.2 Measurement Conditions

The model is verified to measurement data under following bias condition

Gummel Plot:

Table 2-2 Gummel Plot measurement condition of PNP

Gummel Plot			
	Start	Stop	Step
V_{EB}(V)	0.1	1	0.01
V_{CB}(V)	-3.3	-3.3	0

Output characteristics:

Table 2-3 Output characteristic measurement conditions of PNP

Output Characteristics			
	Start	Stop	Step
V_{EC}(V)	0	3.3	0.033
I_B(μA)	-1	-5	-1

Gummel-Poon model has its essential inaccuracy in reverse region, because this model does not have a parameter for the reverse saturation current. Hence, its uses the forward saturation current for both forward and reverse region. In addition, the base resistance parameters (R_B, R_{BM}) were extracted from the forward base current. That is the reason that the reverse base current is not modeled well at high current level.

2.3 PNP_V45X45_L110E Model Verification

2.3.1 PNP_V45X45_L110E model verification at 25 °C

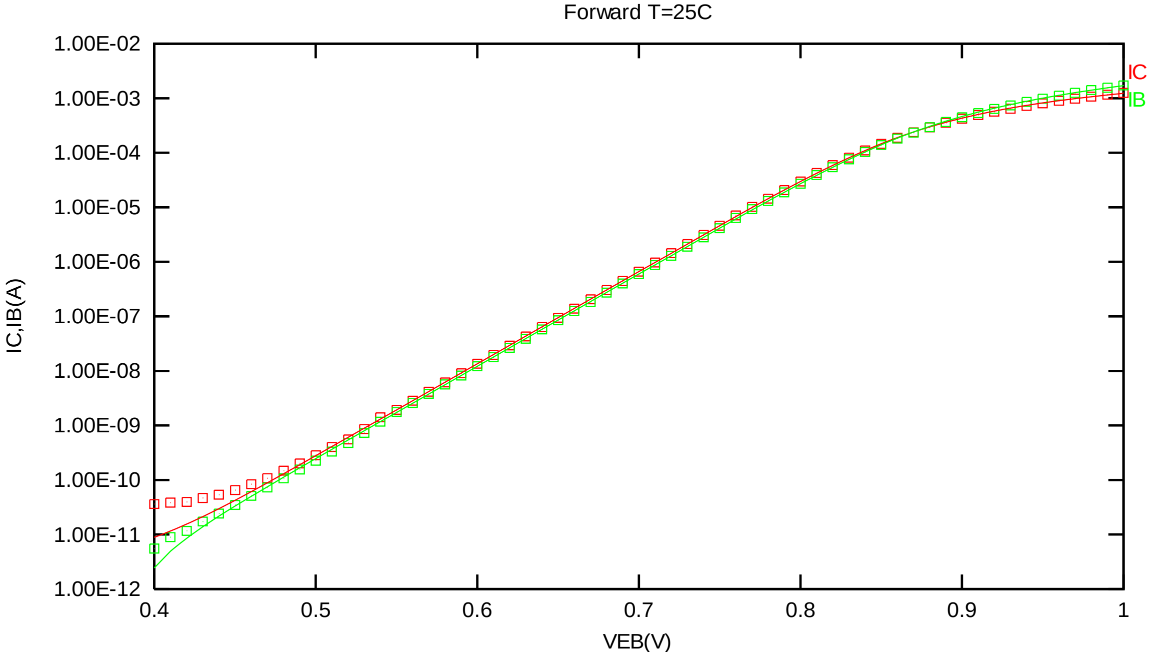

Figure 2-1 Forward gummel plot of PNP_V45X45_L110E at 25 °C, VCB = -3.3V

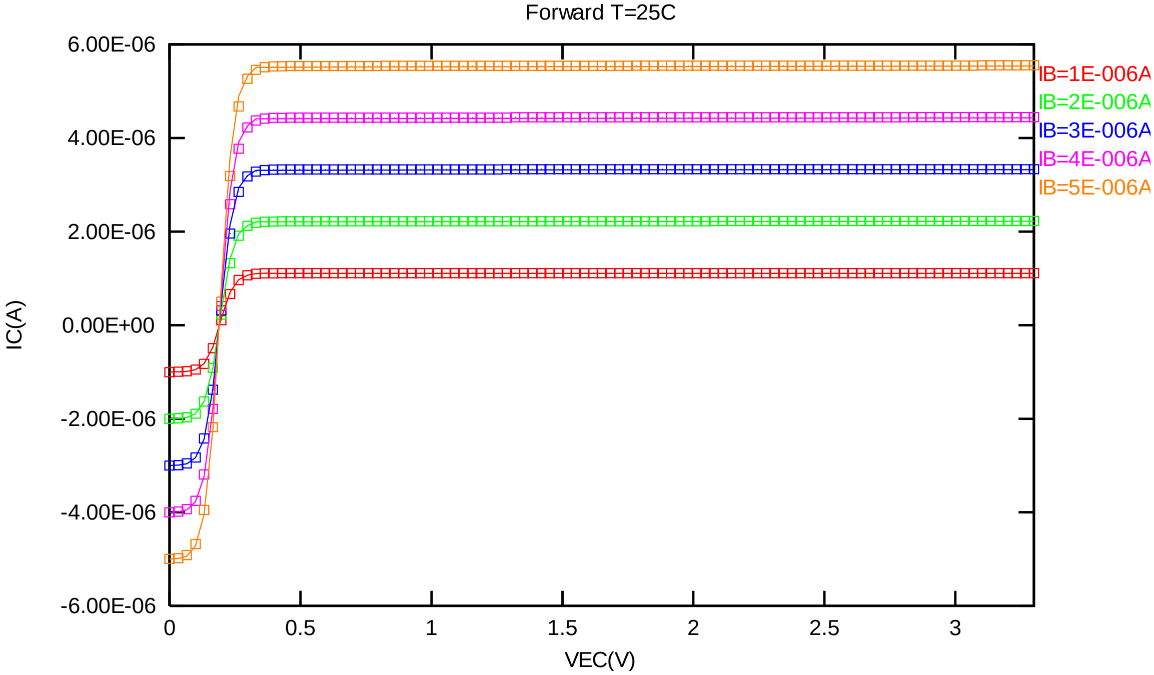

Figure 2-2 Forward output characteristics of PNP_V45X45_L110E at 25 °C

2.3.2 PNP_V45X45_L110E model verification at 85 °C

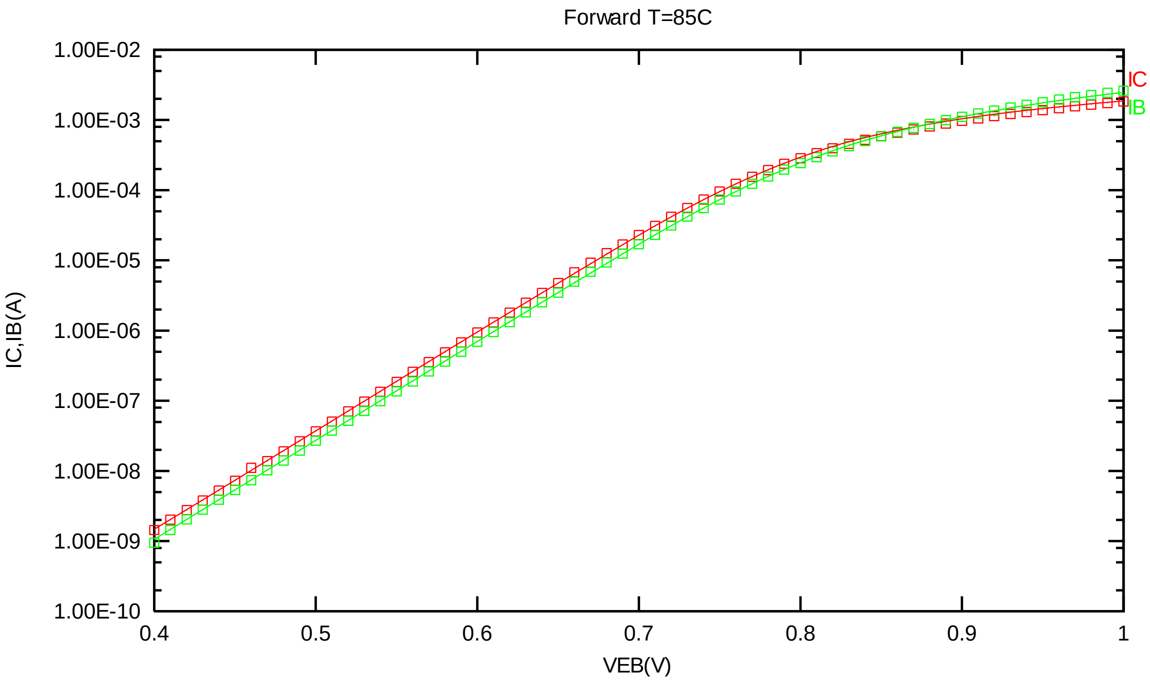

Figure 2-3 Forward gummel plot of PNP_V45X45_L110E at 85 °C, VCB = -3.3V

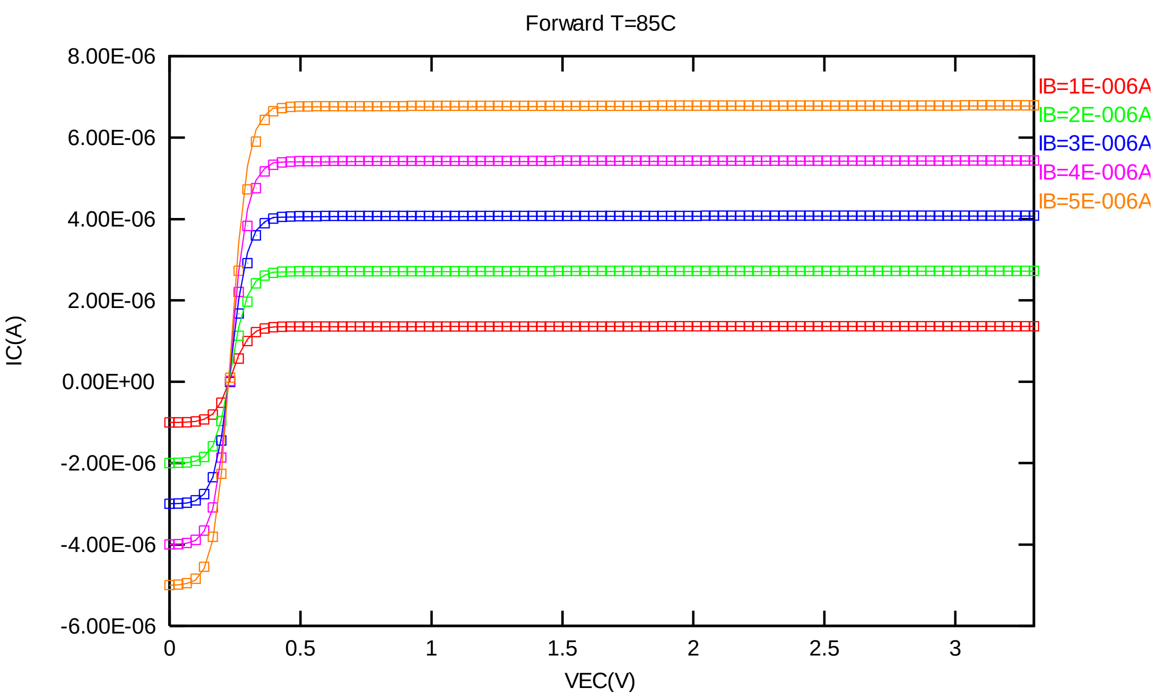

Figure 2-4 Forward output characteristics of PNP_V45X45_L110E at 85 °C

2.3.3 PNP_V45X45_L110E model verification at 125 °C

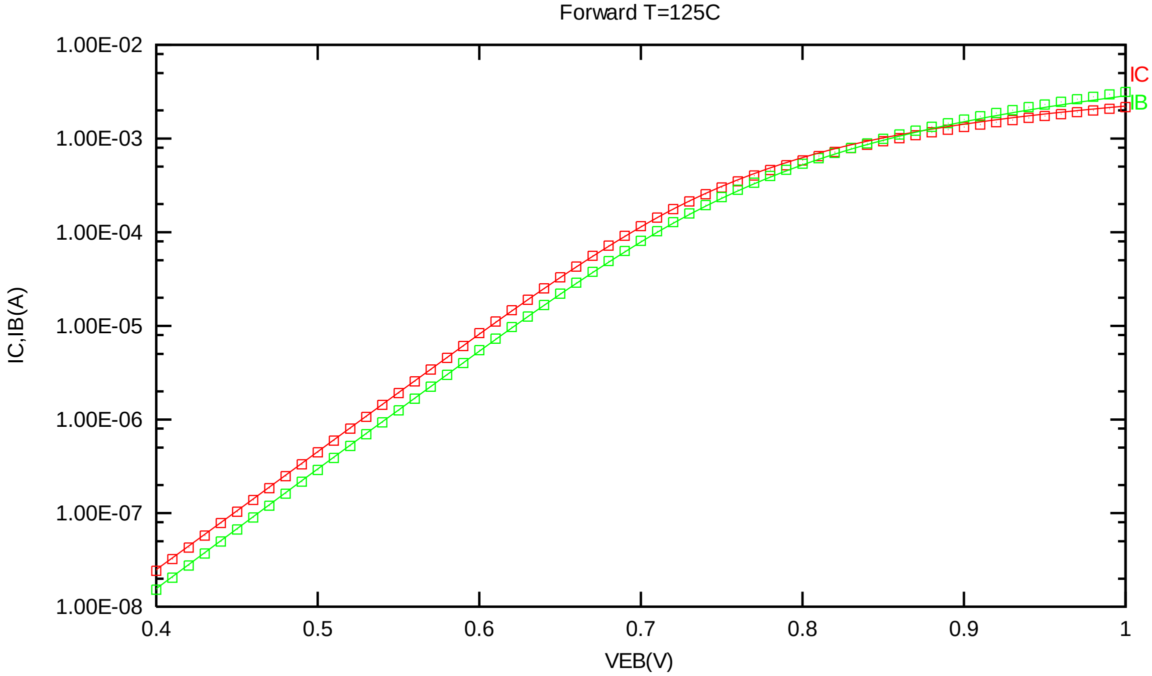

Figure 2-5 Forward gummel plot of PNP_V45X45_L110E at 125 °C, VCB = -3.3V

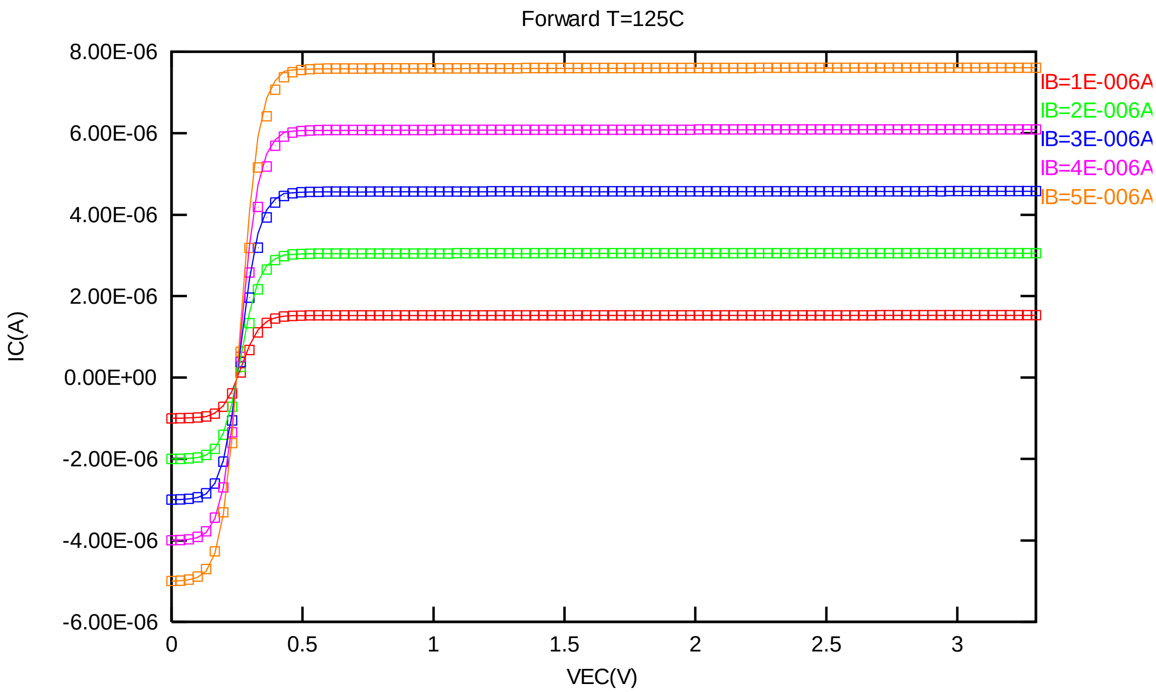

Figure 2-6 Forward output characteristics of PNP_V45X45_L110E at 125 °C

2.3.4 PNP_V45X45_L110E model verification at 0 °C

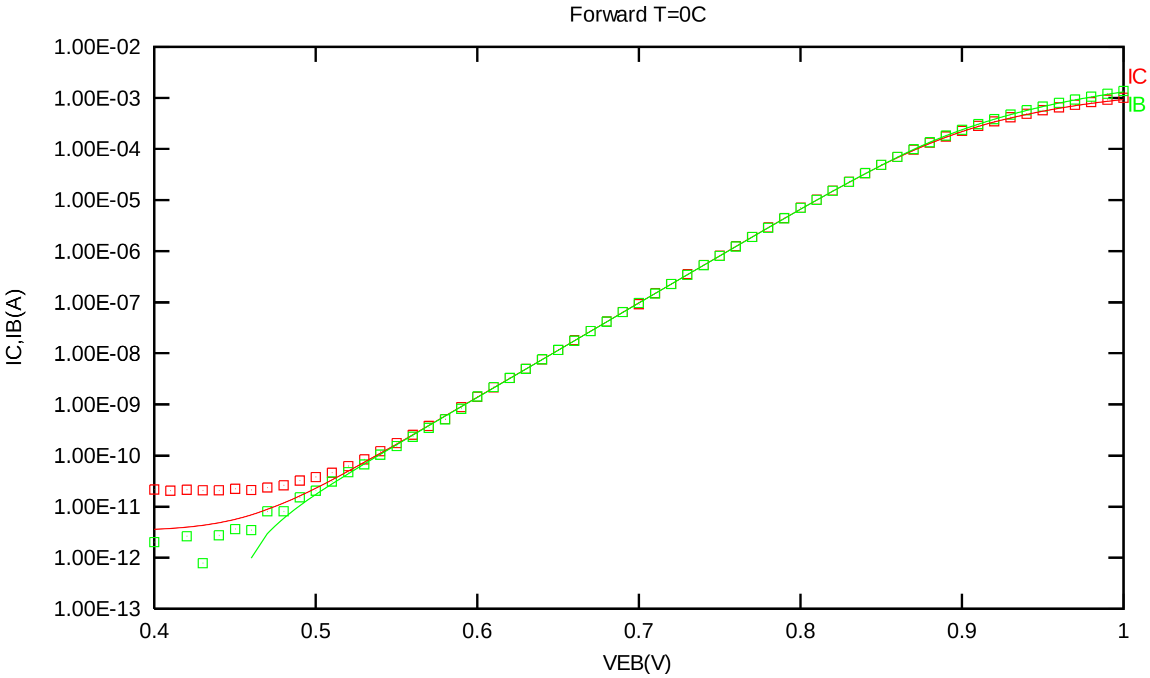

Figure 2-7 Forward gummel plot of PNP_V45X45_L110E at 0 °C, VCB = -3.3V

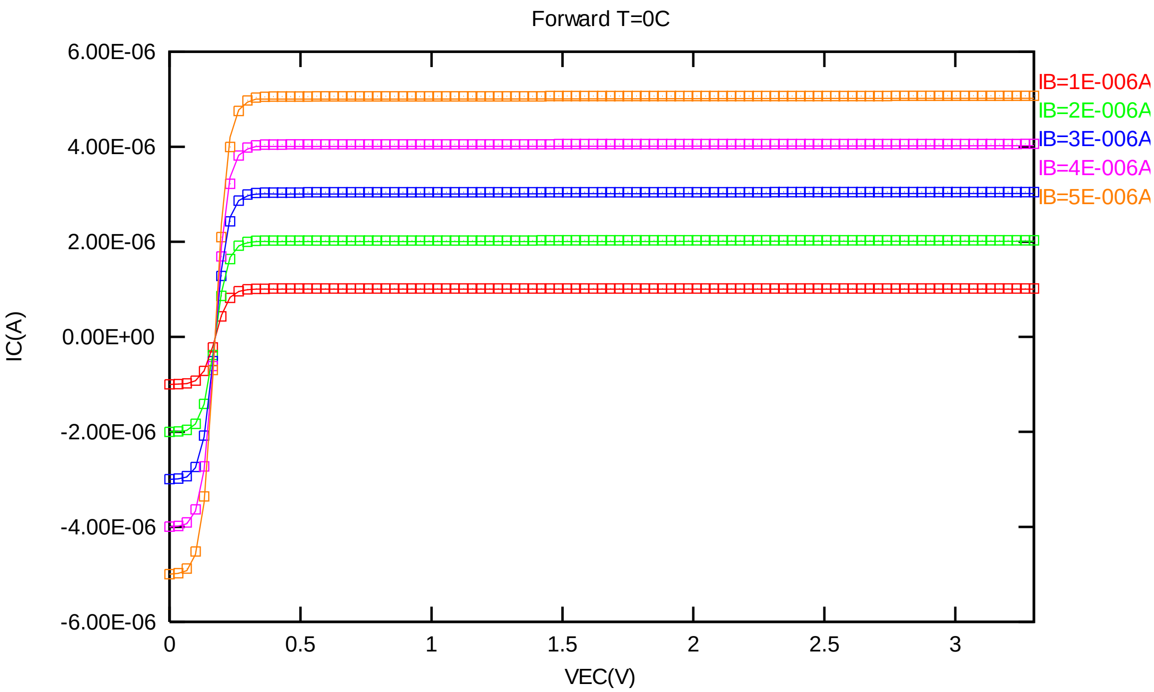

Figure 2-8 Forward output characteristics of PNP_V45X45_L110E at 0 °C

2.3.5 PNP_V45X45_L110E model verification at -50 °C

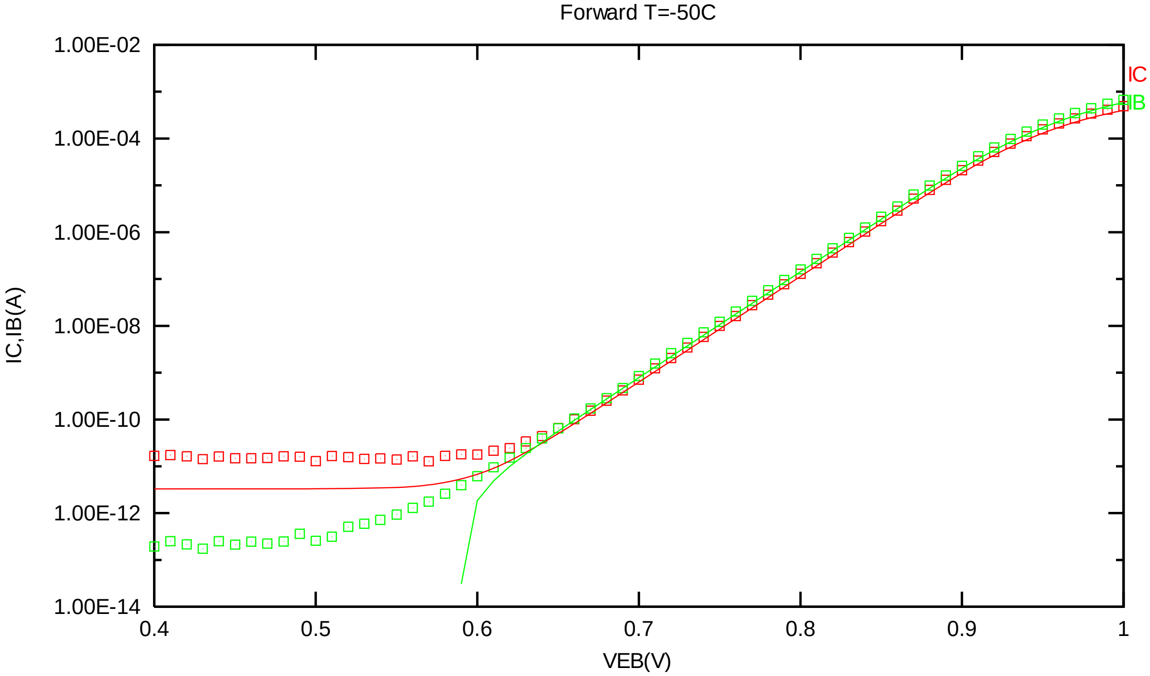

Figure 2-9 Forward gummel plot of PNP_V45X45_L110E at -50 °C, VCB = -3.3V

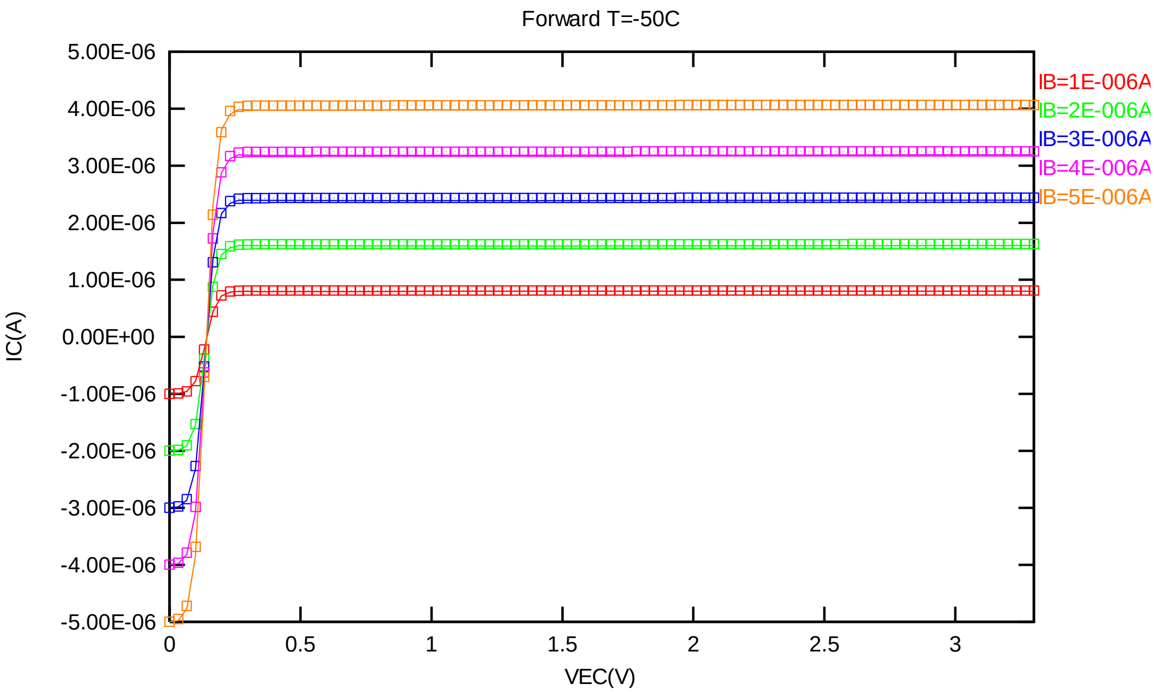

Figure 2-10 Forward output characteristics of PNP_V45X45_L110E at -50 °C

2.3.6 PNP_V45X45_L110E PNP model temperature effect

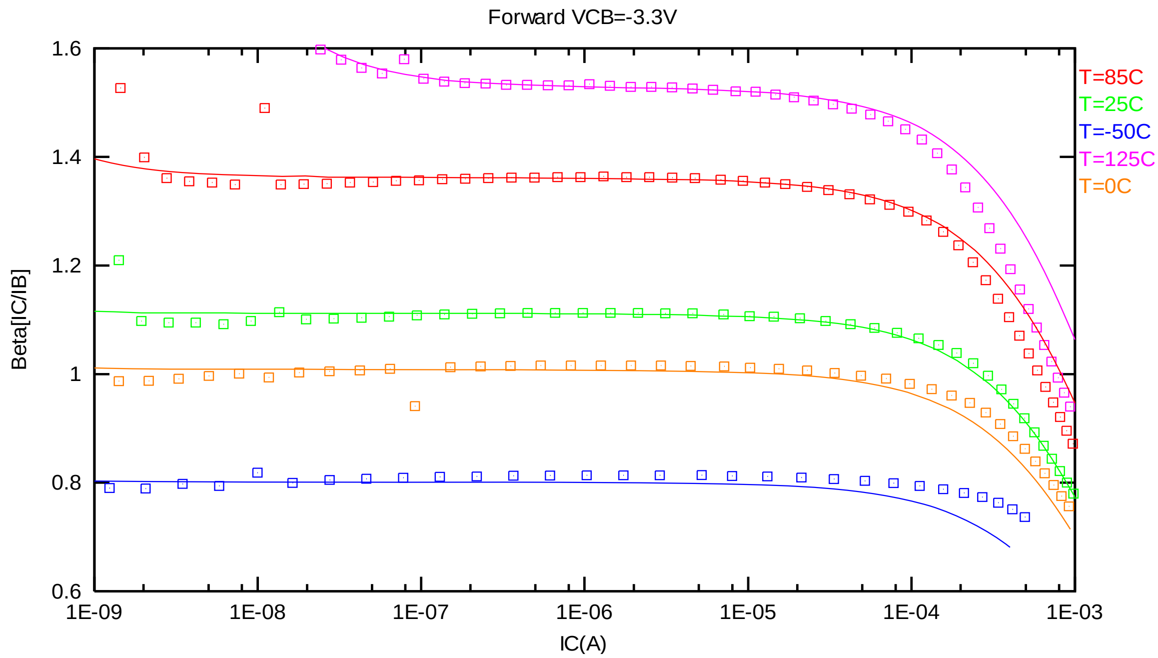

Figure 2-11 Current gain vs. IC at different temperatures of PNP_V45X45_L110E, VCB = -3.3V

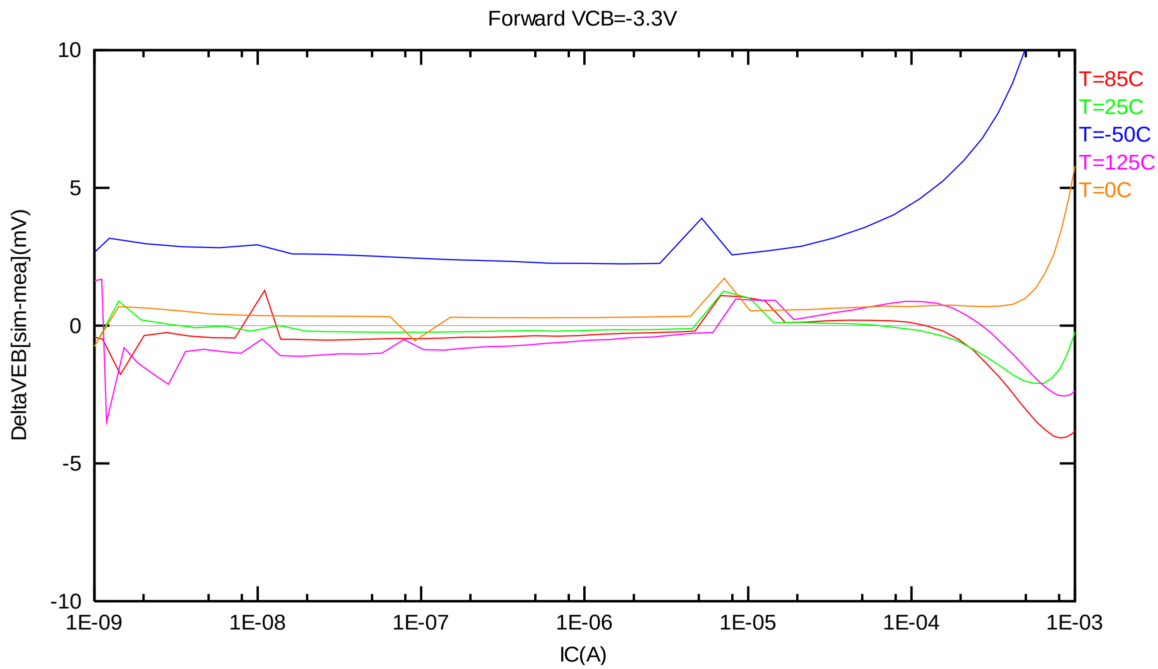

Figure 2-12 DeltaVEB vs. IC at different temperatures of PNP_V45X45_L110E, VCB = -3.3V

2.4 PNP_V90X90_L110E Model Verification

2.4.1 PNP_V90X90_L110E model verification at 25 °C

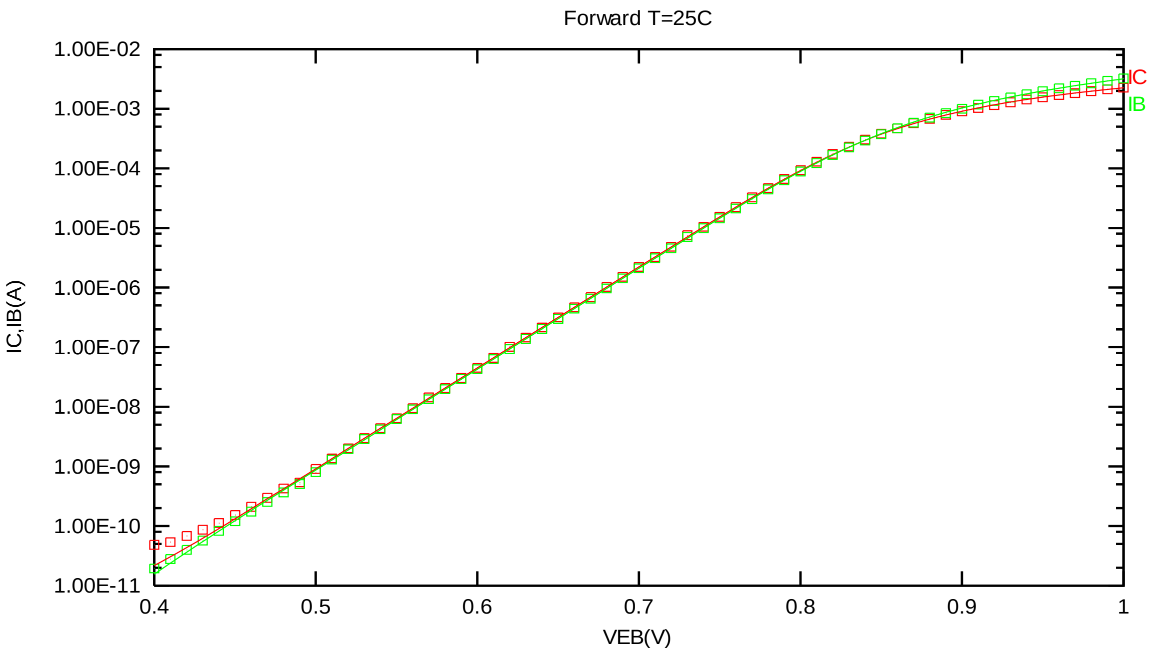

Figure 2-13 Forward gummel plot of PNP_V90X90_L110E at 25 °C, VCB = -3.3V

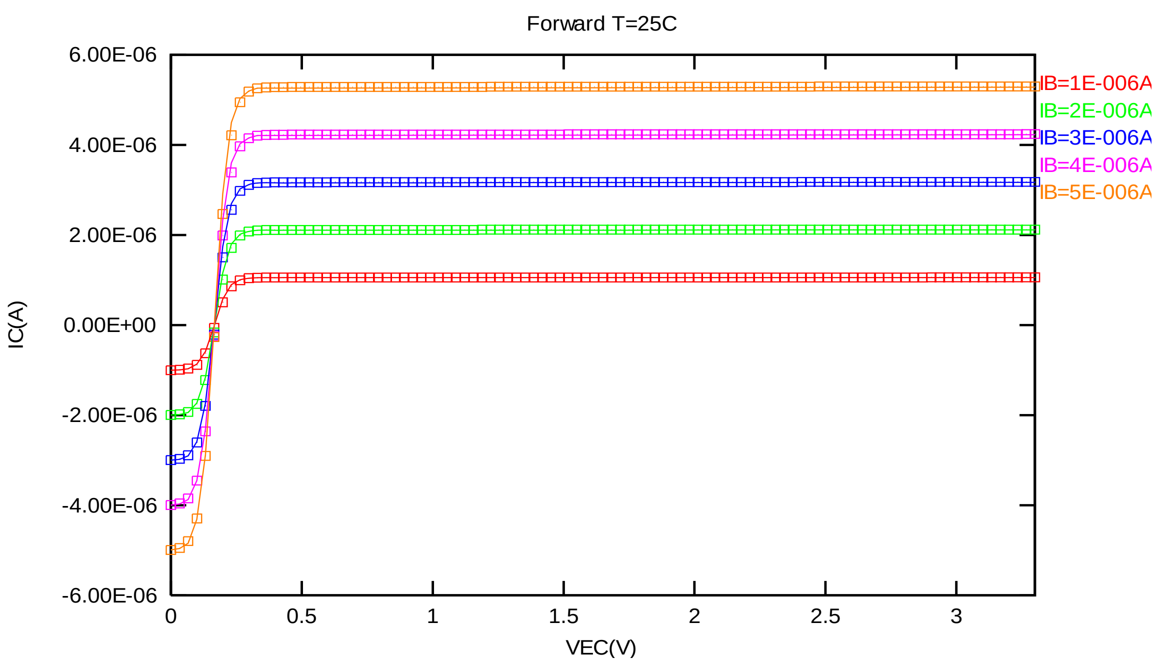

Figure 2-14 Forward output characteristics of PNP_V90X90_L110E at 25 °C

2.4.2 PNP_V90X90_L110E model verification at 85 °C

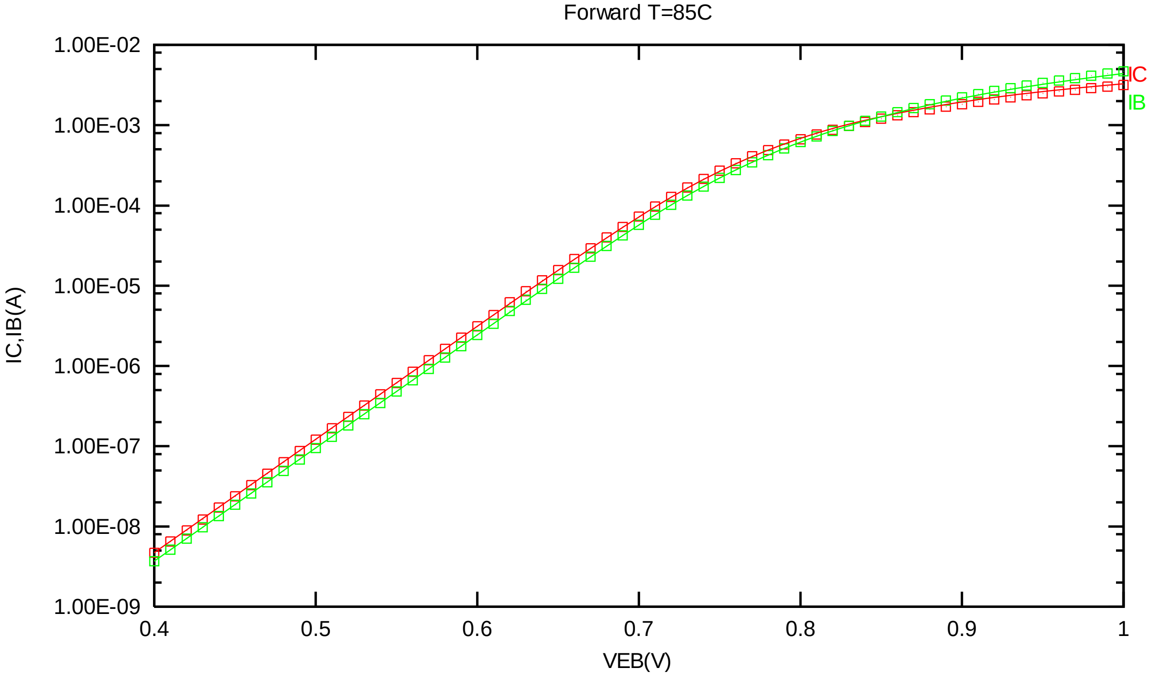

Figure 2-15 Forward gummel plot of PNP_V90X90_L110E at 85 °C, VCB = -3.3V

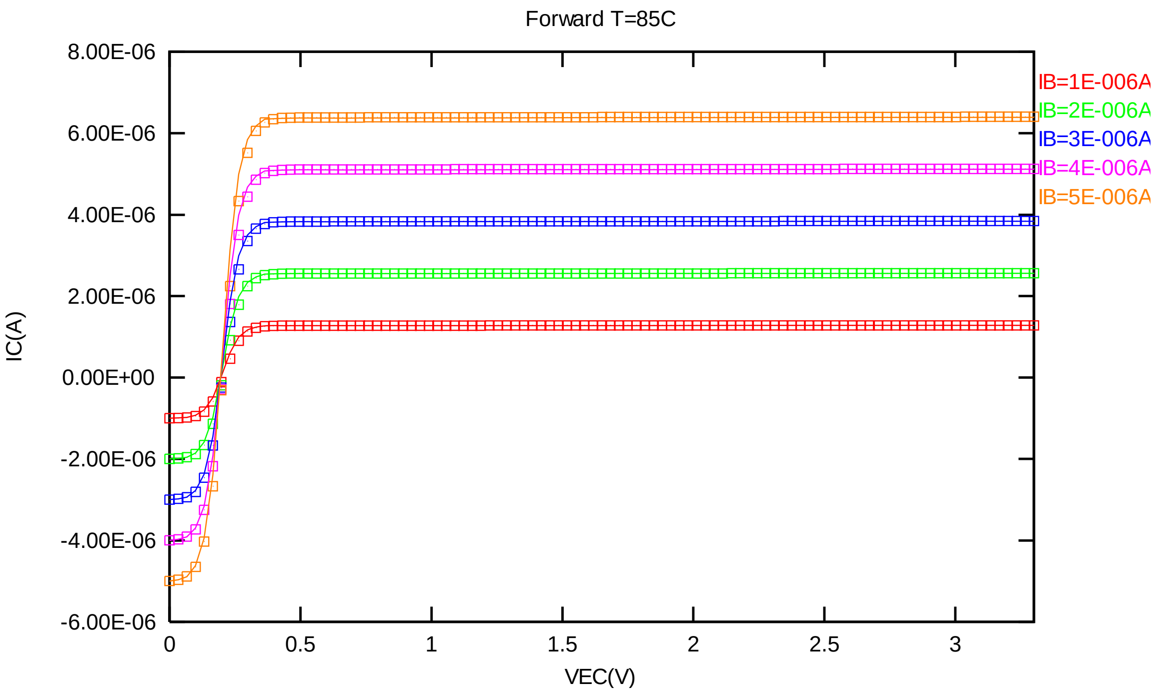

Figure 2-16 Forward output characteristics of PNP_V90X90_L110E at 85 °C

2.4.3 PNP_V90X90_L110E model verification at 125 °C

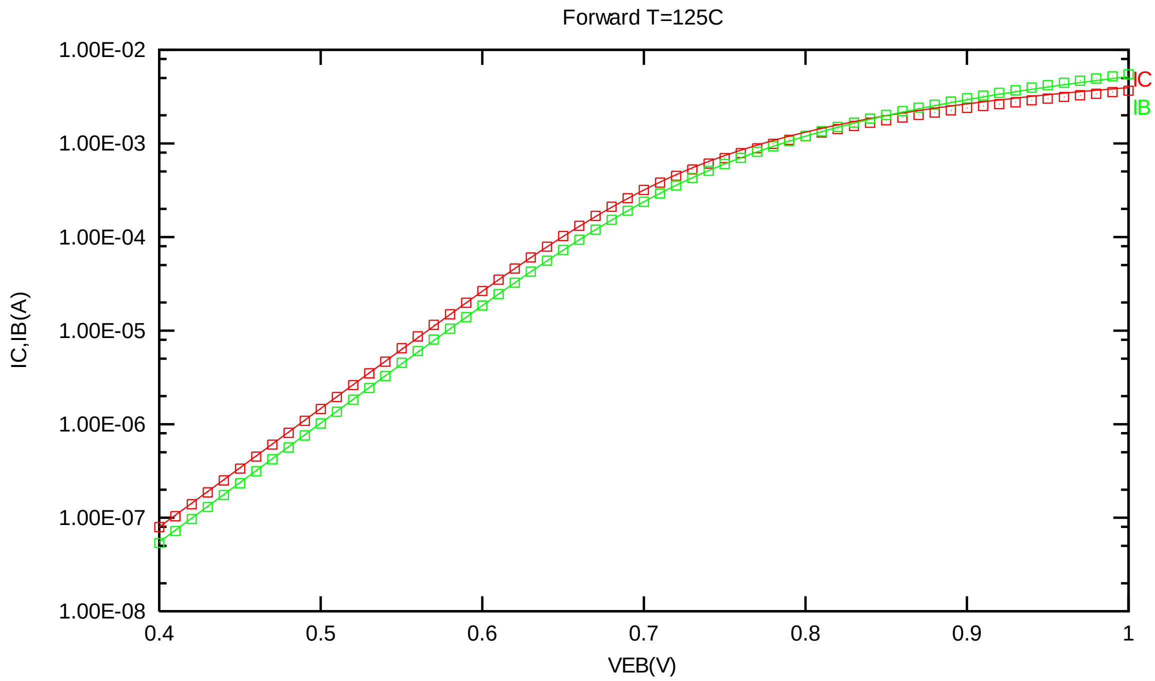

Figure 2-17 Forward gummel plot of PNP_V90X90_L110E at 125 °C, VCB = -3.3V

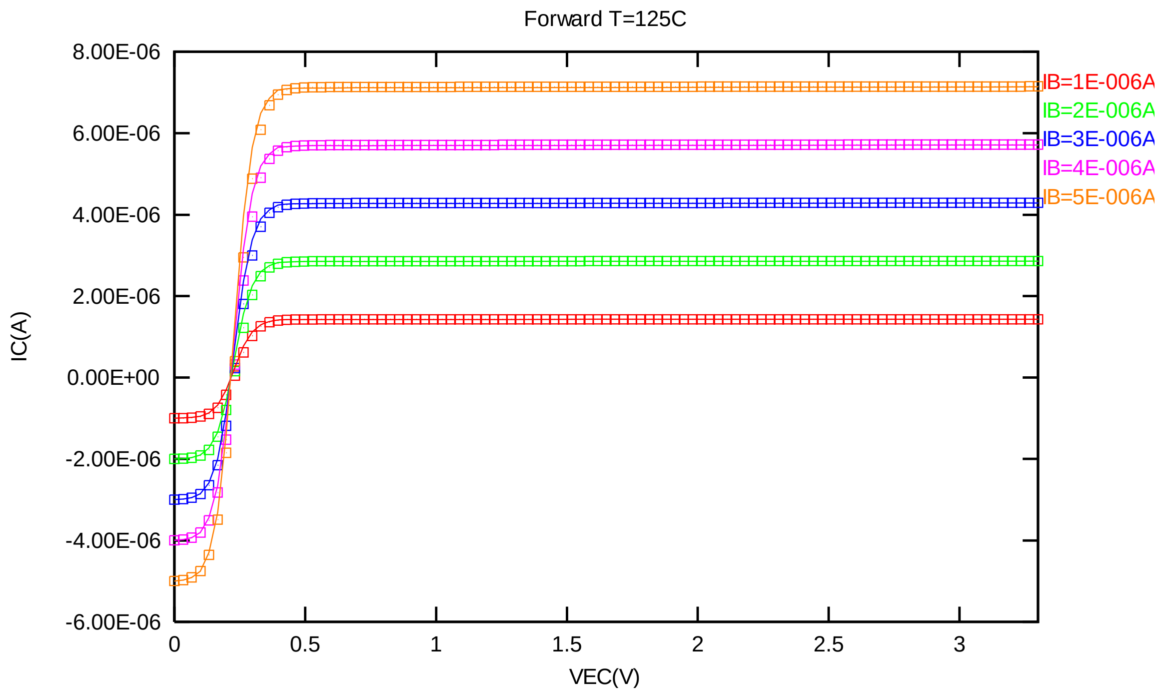

Figure 2-18 Forward output characteristics of PNP_V90X90_L110E at 125 °C

2.4.4 PNP_V90X90_L110E model verification at 0 °C

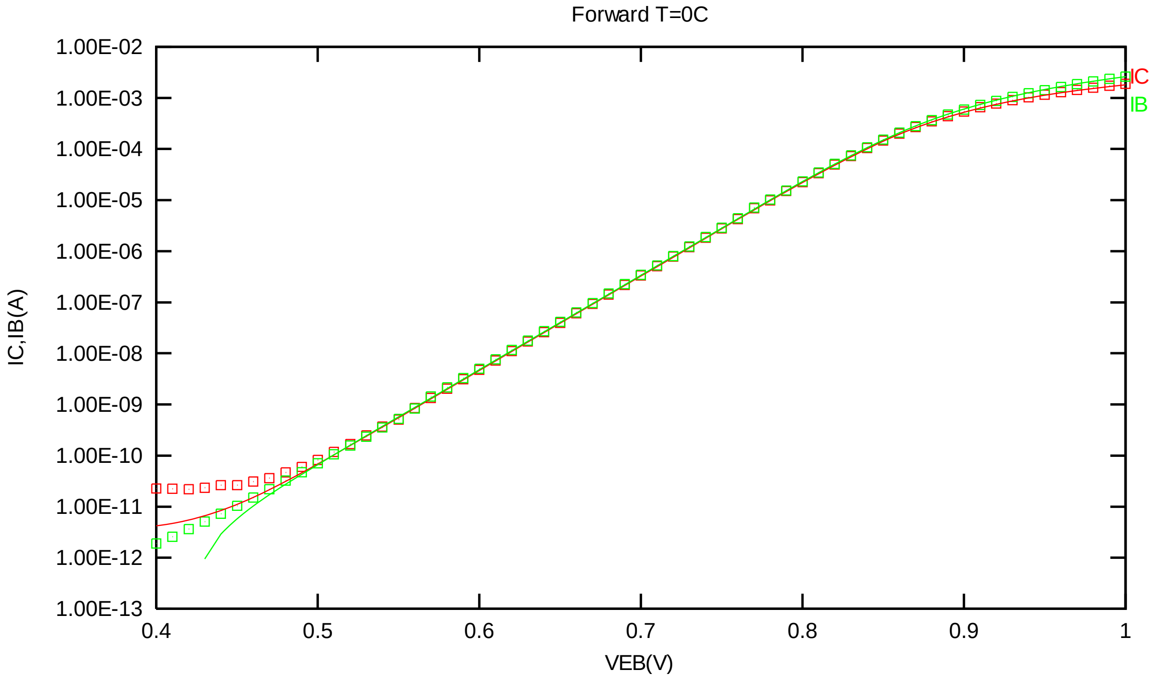

Figure 2-19 Forward gummel plot of PNP_V90X90_L110E at 0 °C, VCB = -3.3V

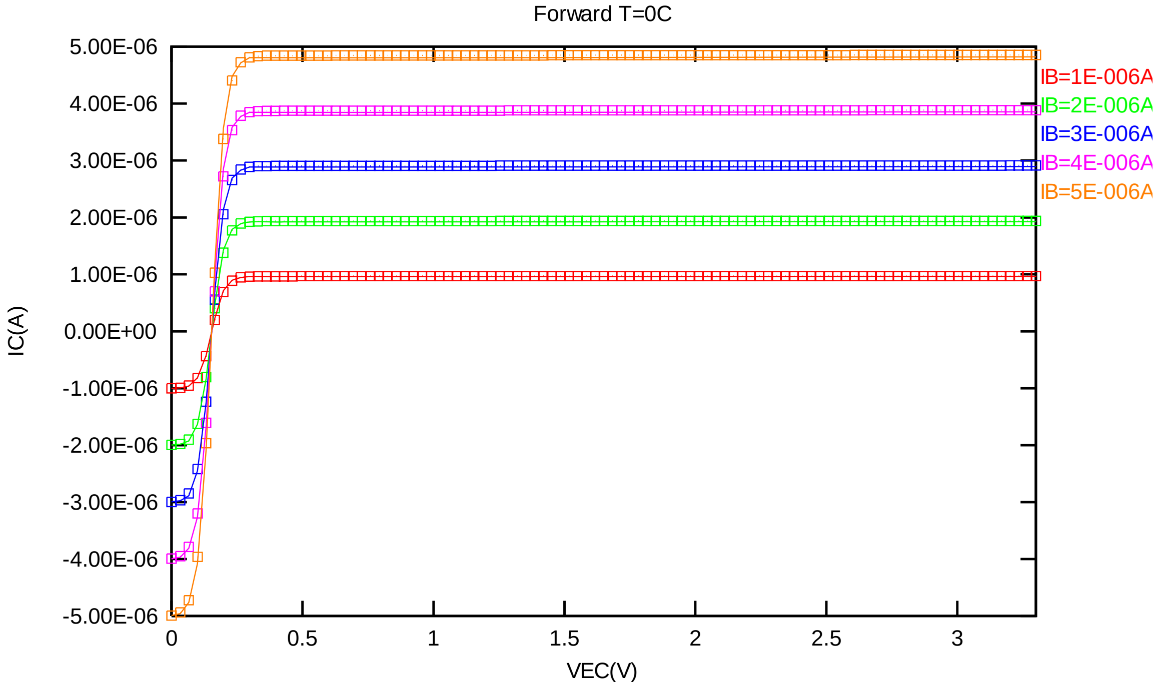

Figure 2-20 Forward output characteristics of PNP_V90X90_L110E at 0 °C

2.4.5 PNP_V90X90_L110E model verification at -50 °C

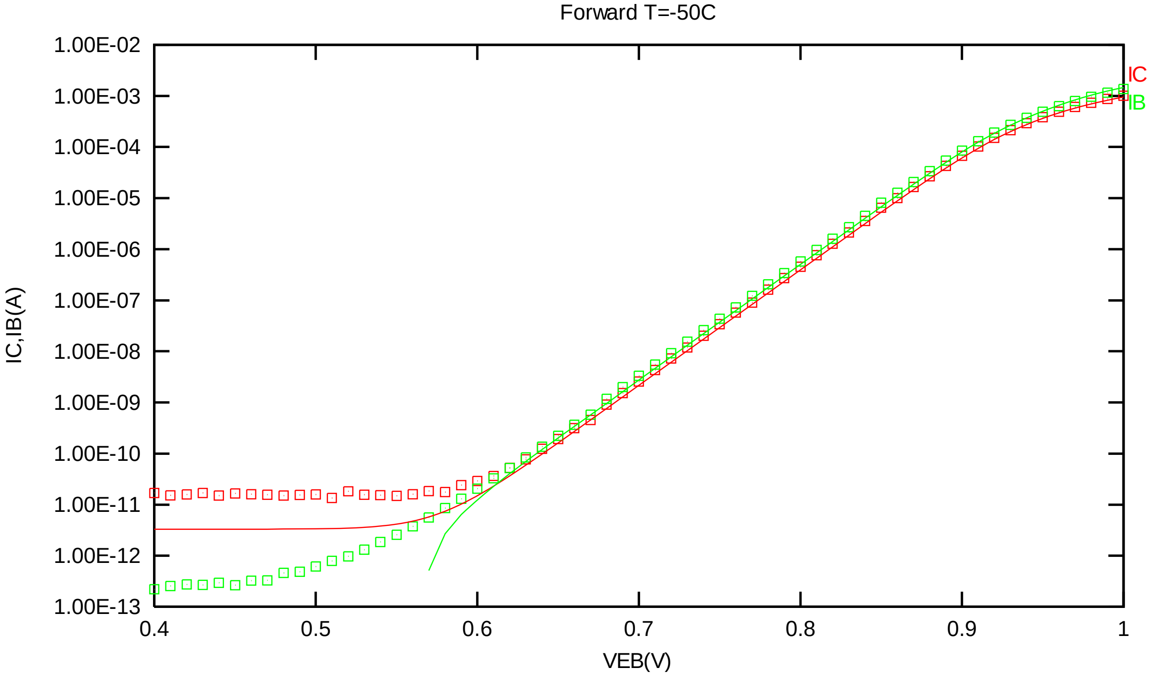

Figure 2-21 Forward gummel plot of PNP_V90X90_L110E at -50 °C, VCB = -3.3V

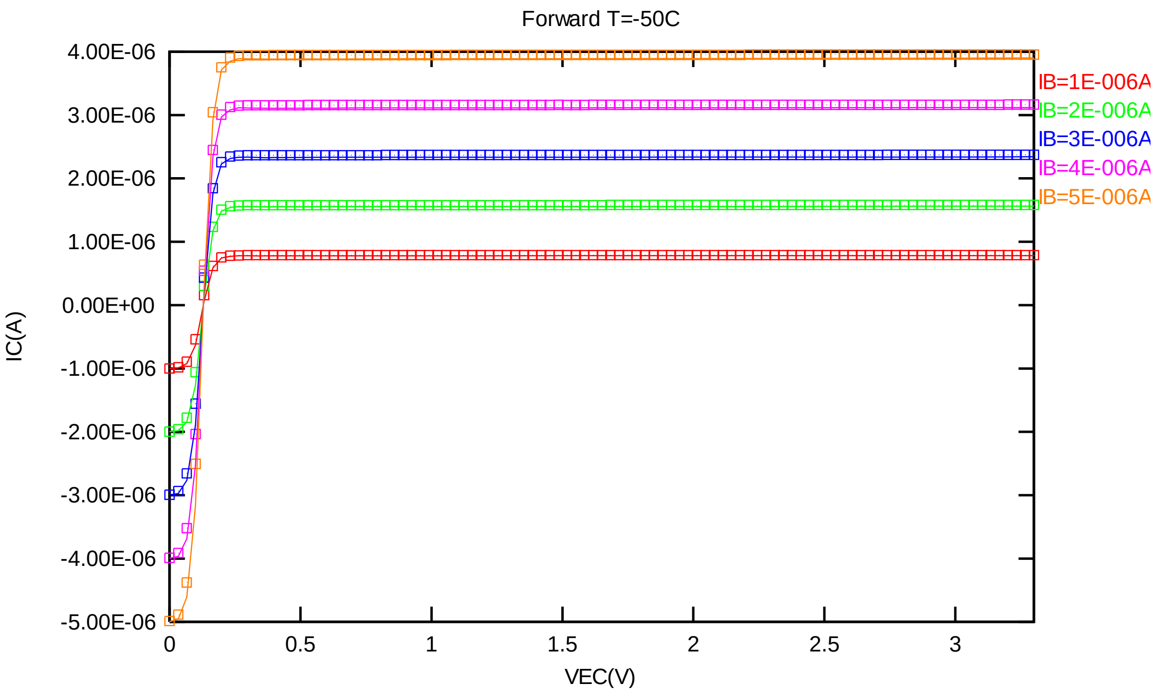

Figure 2-22 Forward output characteristics of PNP_V90X90_L110E at -50 °C

2.4.6 PNP_V90X90_L110E PNP model temperature effect

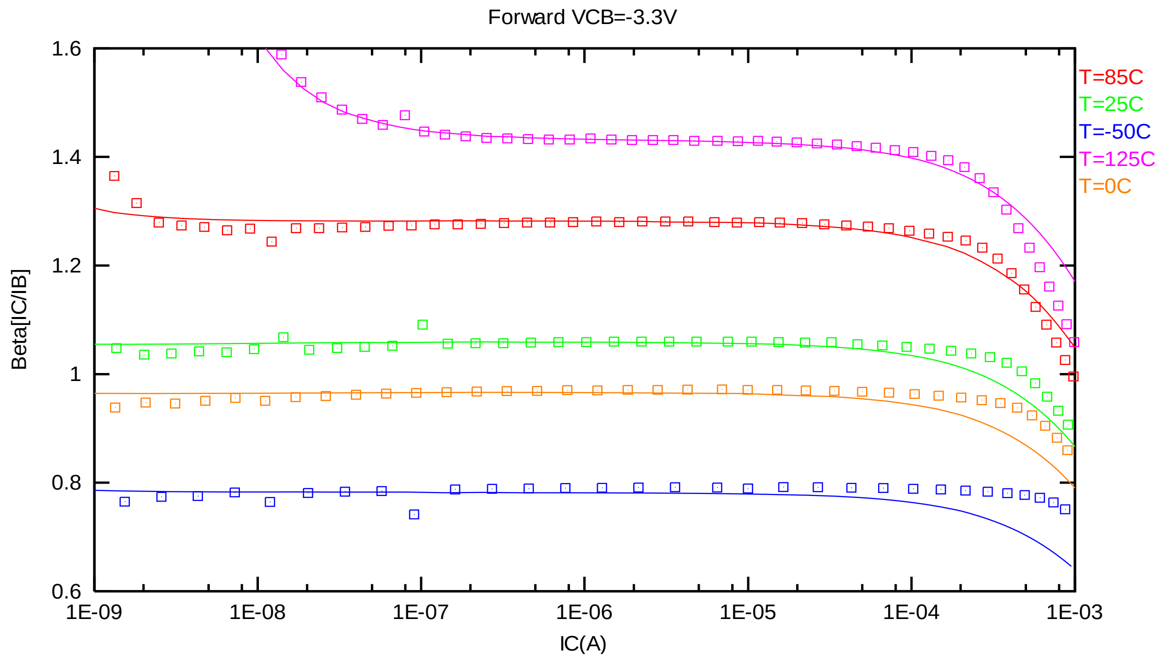

Figure 2-23 Current gain vs. IC at different temperatures of PNP_V90X90_L110E, VCB = -3.3V

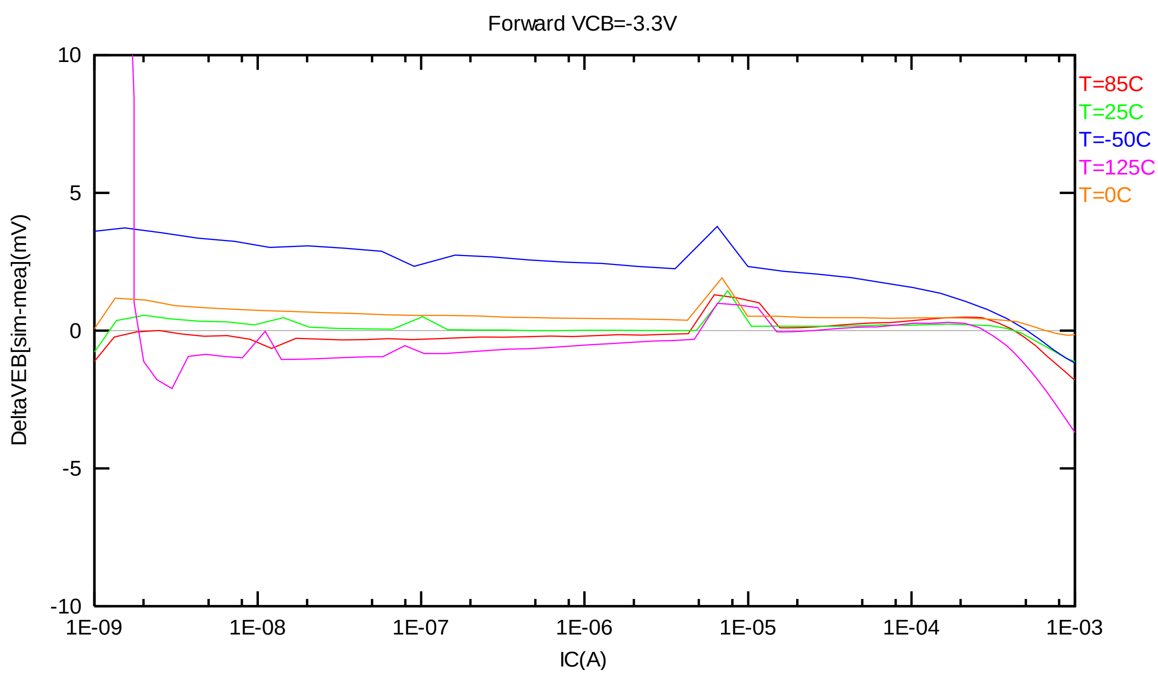

Figure 2-24 DeltaVEB vs. IC at different temperatures of PNP_V90X90_L110E, VCB = -3.3V

2.5 PNP_V135X135_L110E Model Verification

2.5.1 PNP_V135X135_L110E model verification at 25 °C

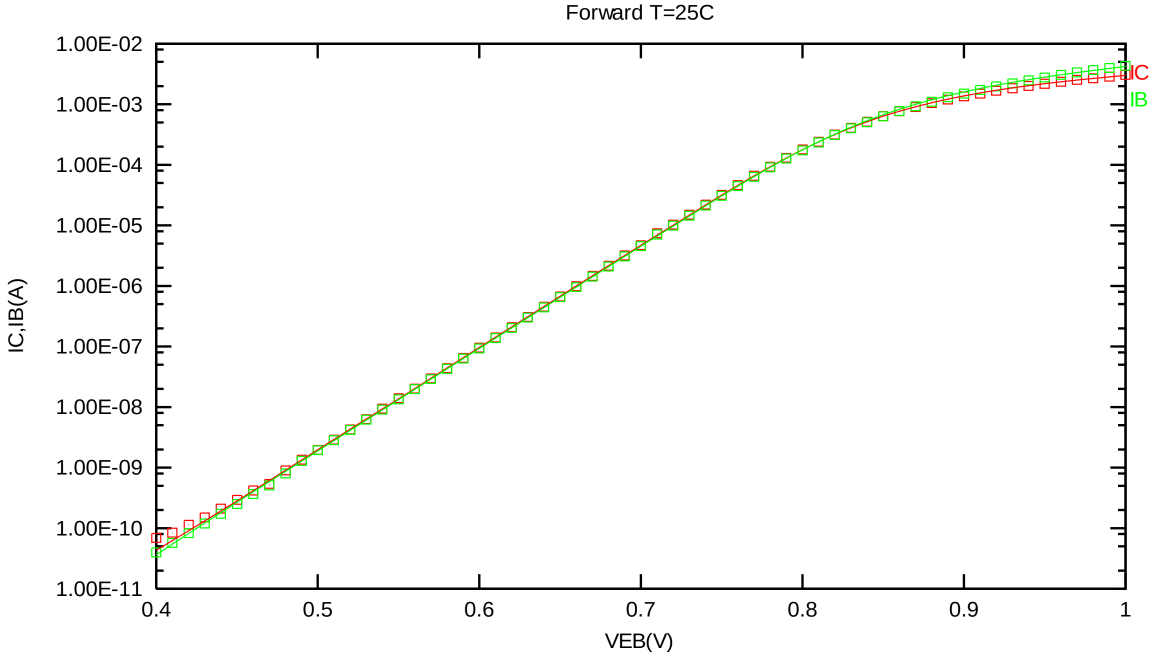

Figure 2-25 Forward gummel plot of PNP_V135X135_L110E at 25 °C, VCB = -3.3V

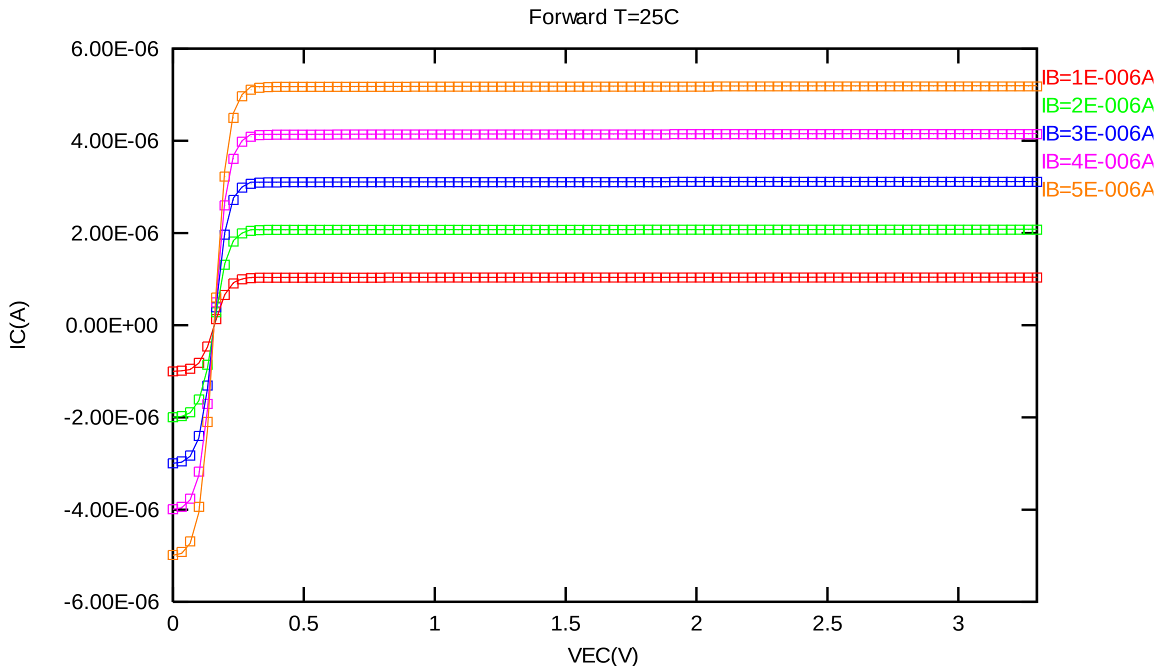

Figure 2-26 Forward output characteristics of PNP_V135X135_L110E at 25 °C

2.5.2 PNP_V135X135_L110E model verification at 85 °C

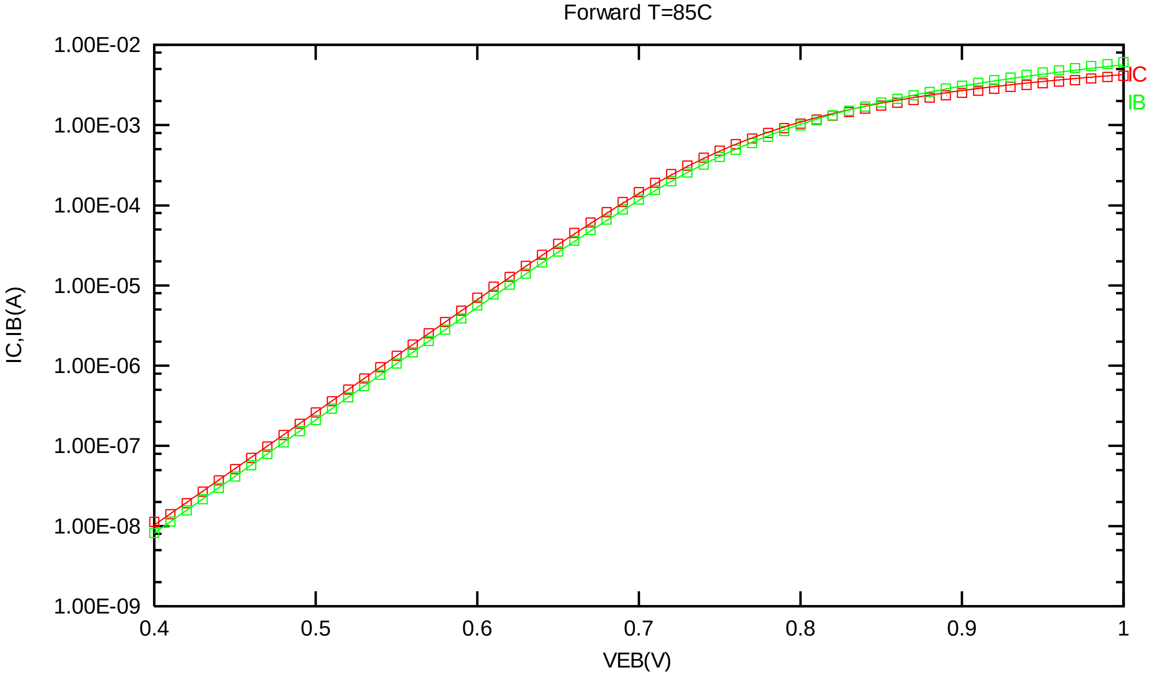

Figure 2-27 Forward gummel plot of PNP_V135X135_L110E at 85 °C, VCB = -3.3V

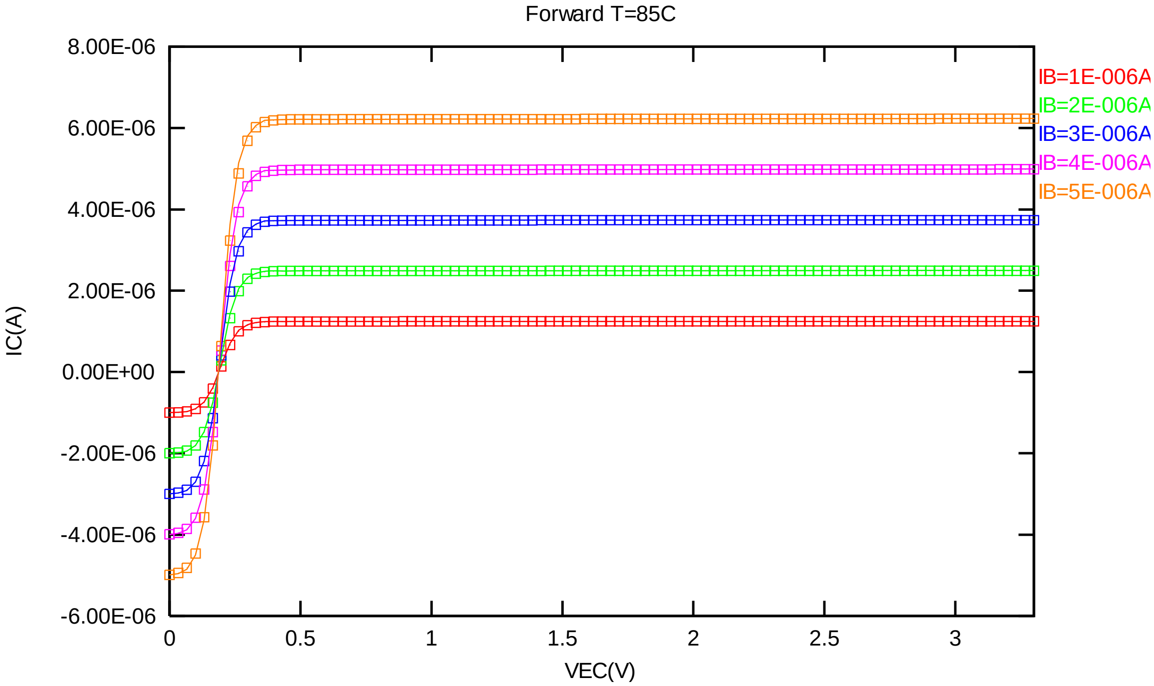

Figure 2-28 Forward output characteristics of PNP_V135X135_L110E at 85 °C

2.5.3 PNP_V135X135_L110E model verification at 125 °C

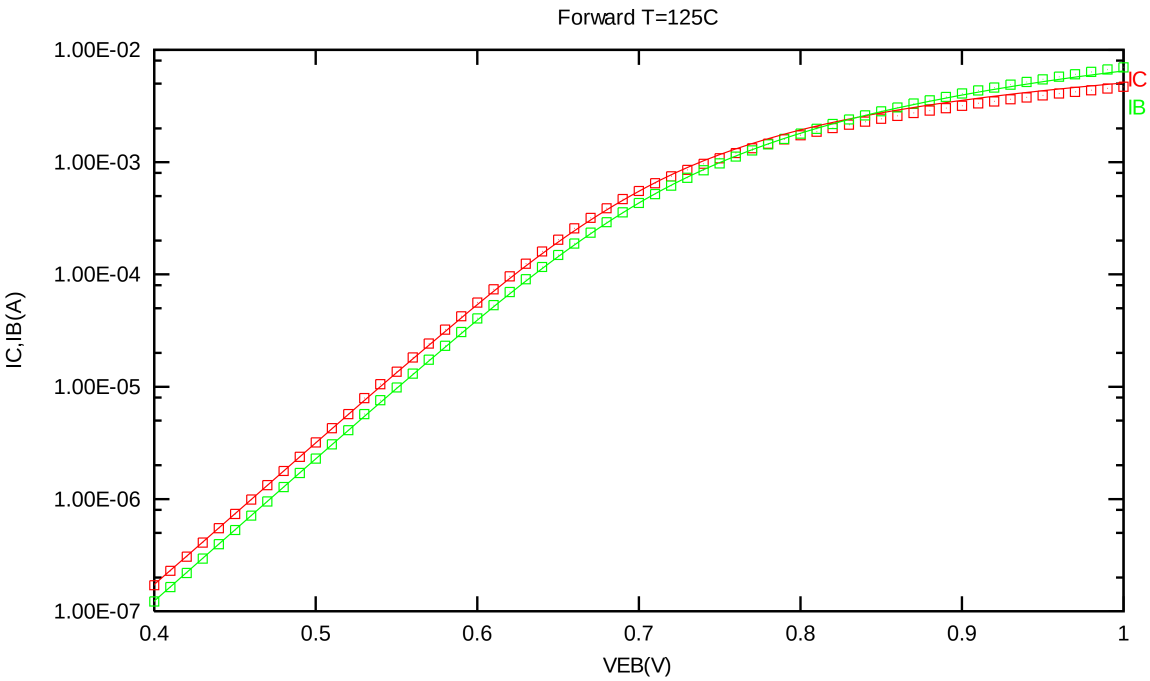

Figure 2-29 Forward gummel plot of PNP_V135X135_L110E at 125 °C, VCB = -3.3V

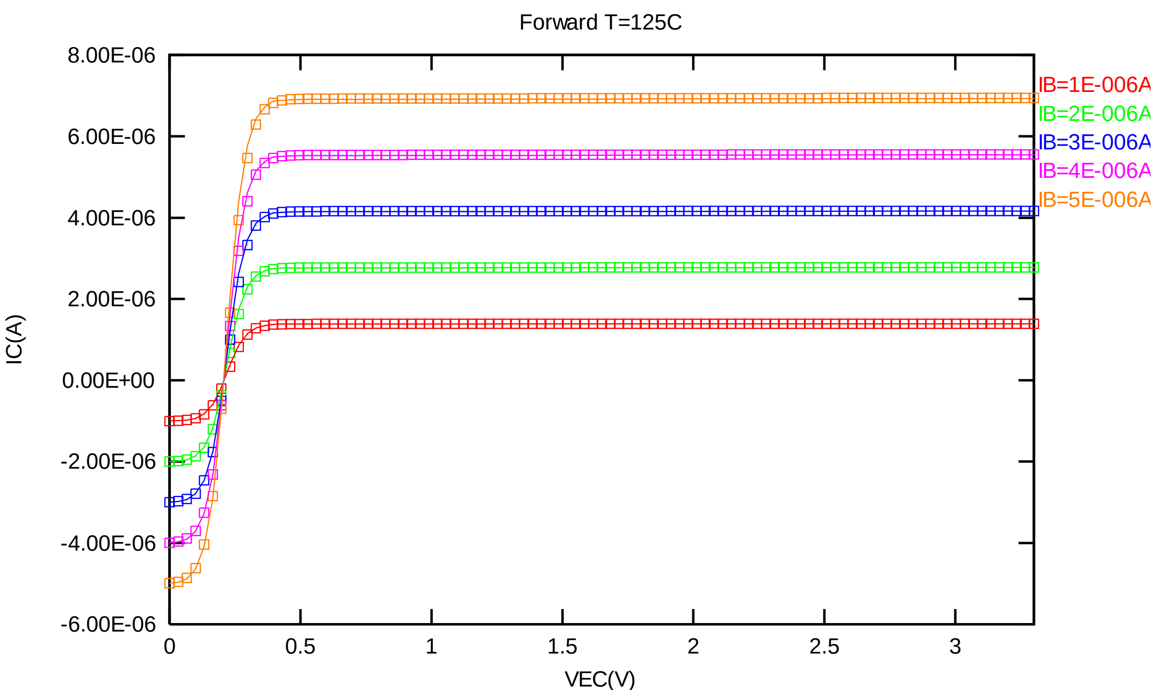

Figure 2-30 Forward output characteristics of PNP_V135X135_L110E at 125 °C

2.5.4 PNP_V135X135_L110E model verification at 0 °C

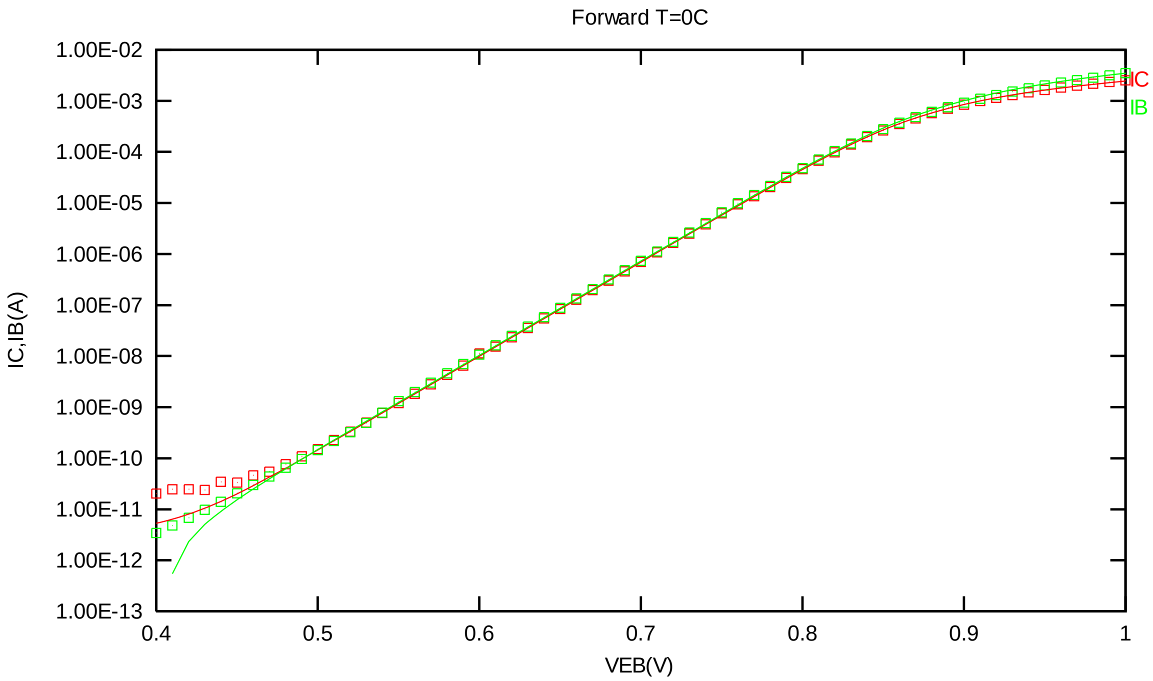

Figure 2-31 Forward gummel plot of PNP_V135X135_L110E at 0 °C, VCB = -3.3V

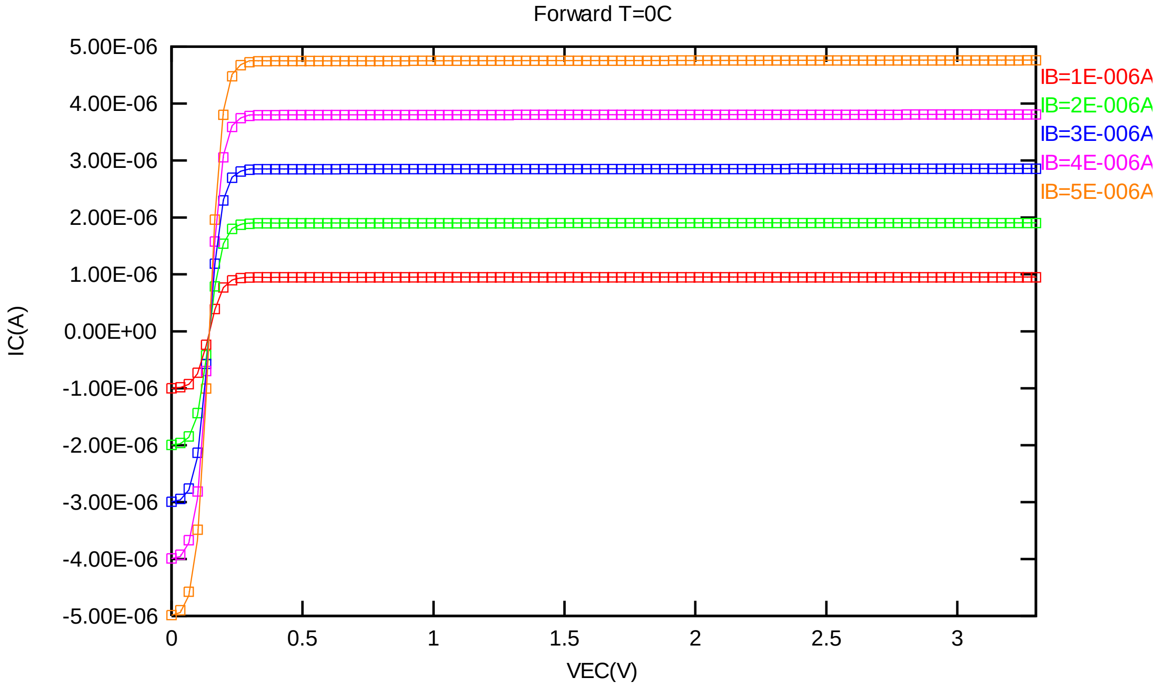

Figure 2-32 Forward output characteristics of PNP_V135X135_L110E at 0 °C

2.5.5 PNP_V135X135_L110E model verification at -50 °C

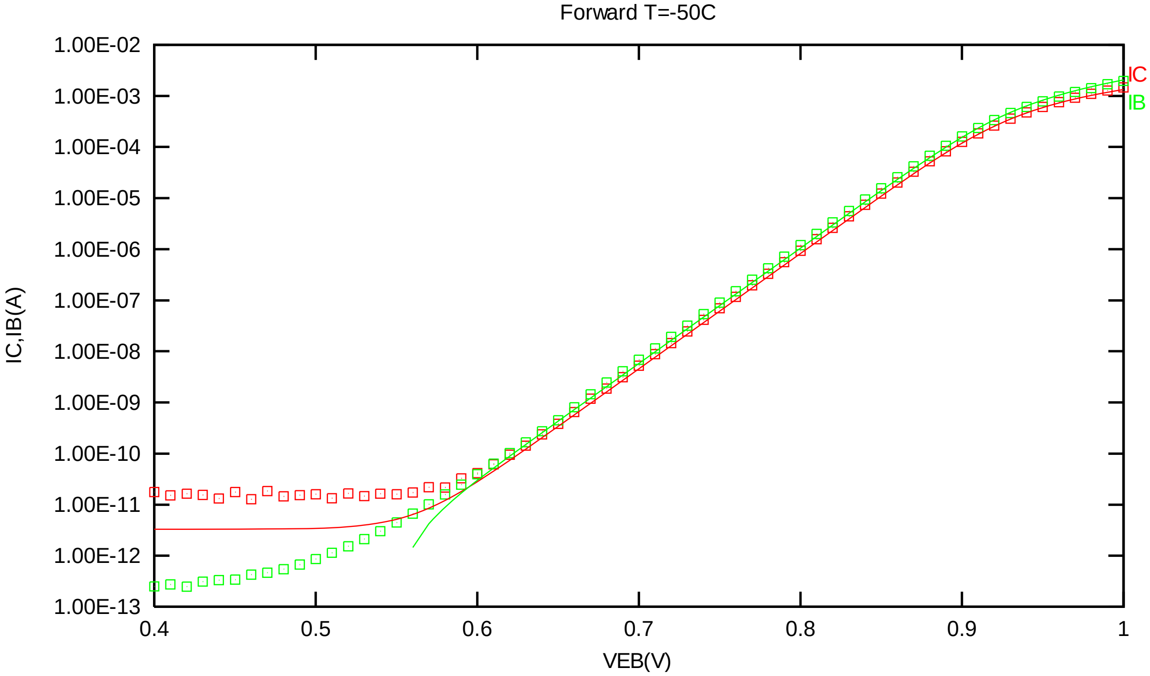

Figure 2-33 Forward gummel plot of PNP_V135X135_L110E at -50 °C, VCB = -3.3V

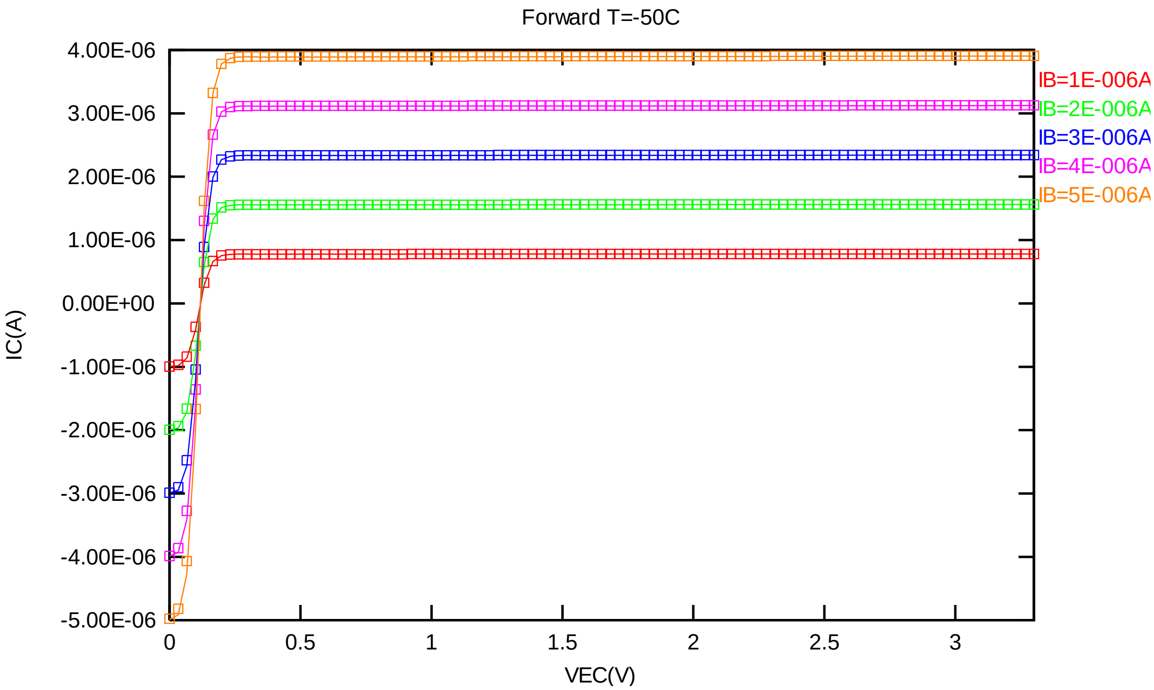

Figure 2-34 Forward output characteristics of PNP_V135X135_L110E at -50 °C

2.5.6 PNP_V135X135_L110E PNP model temperature effect

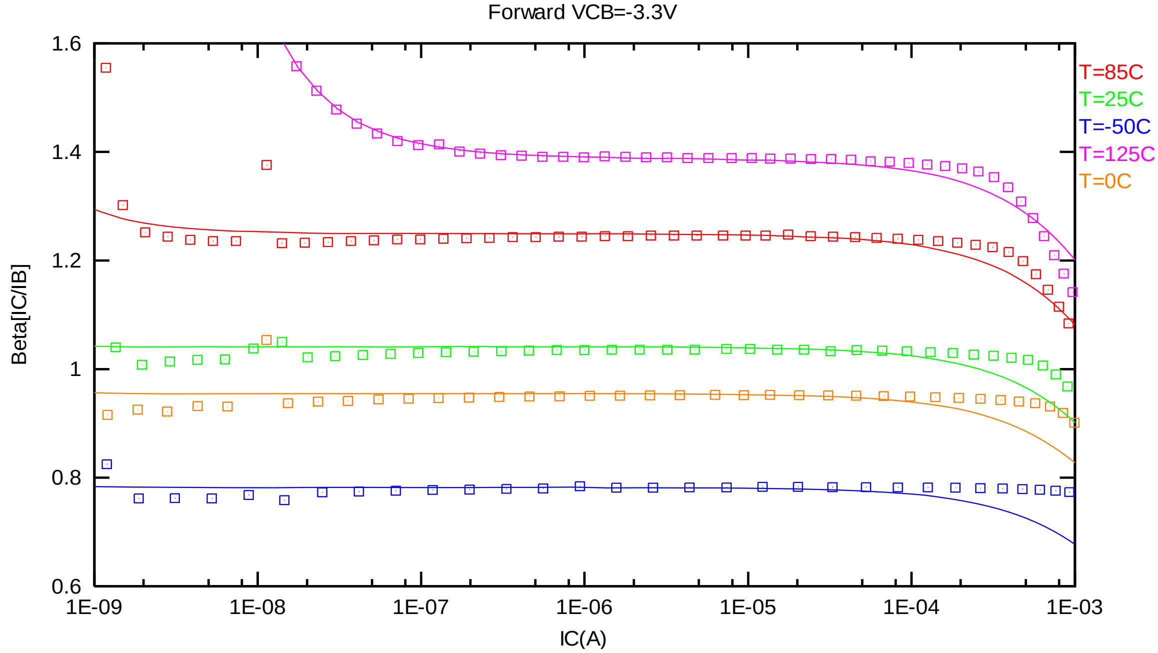

Figure 2-35 Current gain vs. IC at different temperatures of PNP_V135X135_L110E, VCB = -3.3V

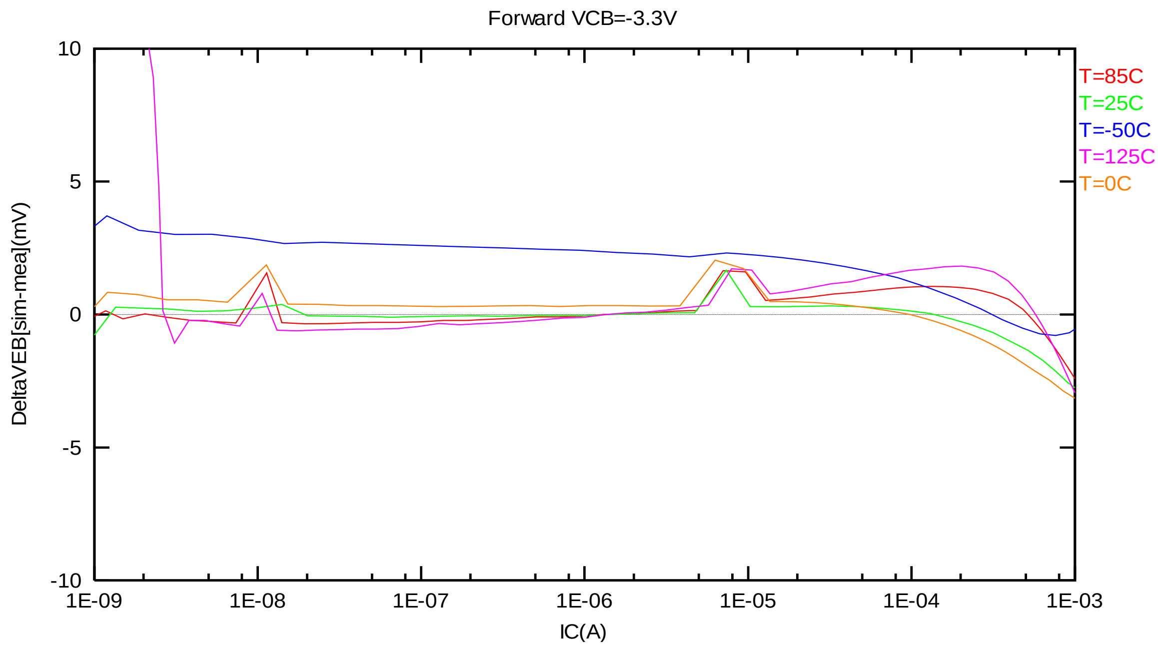

Figure 2-36 DeltaVEB vs. IC at different temperatures of PNP_V135X135_L110E, VCB = -3.3V

2.6 PNP_SV45X45_L110E Model Verification

2.6.1 PNP_SV45X45_L110E model verification at 25 °C

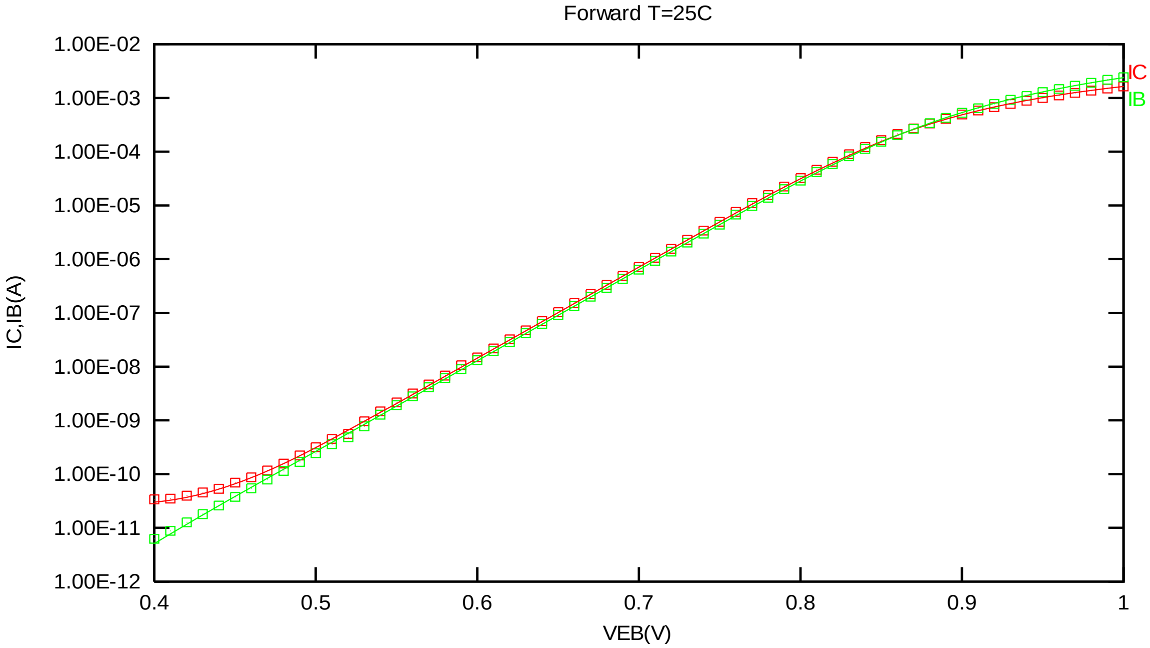

Figure 2-37 Forward gummel plot of PNP_SV45X45_L110E at 25 °C, VCB = -3.3V

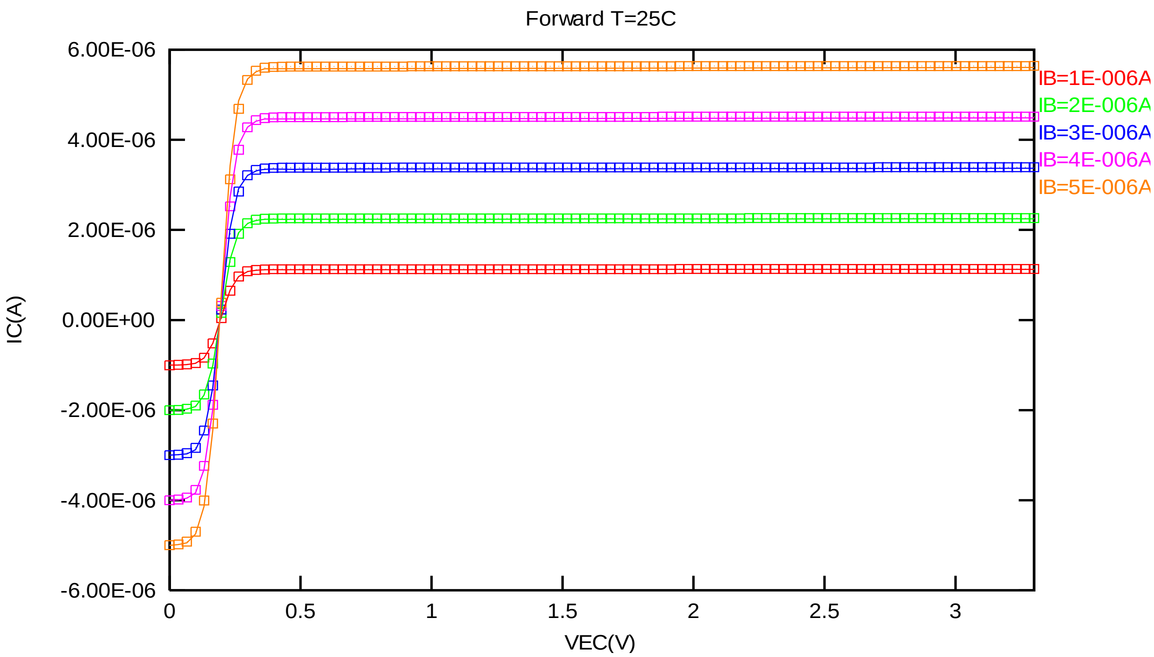

Figure 2-38 Forward output characteristics of PNP_SV45X45_L110E at 25 °C

2.6.2 PNP_SV45X45_L110E model verification at 85 °C

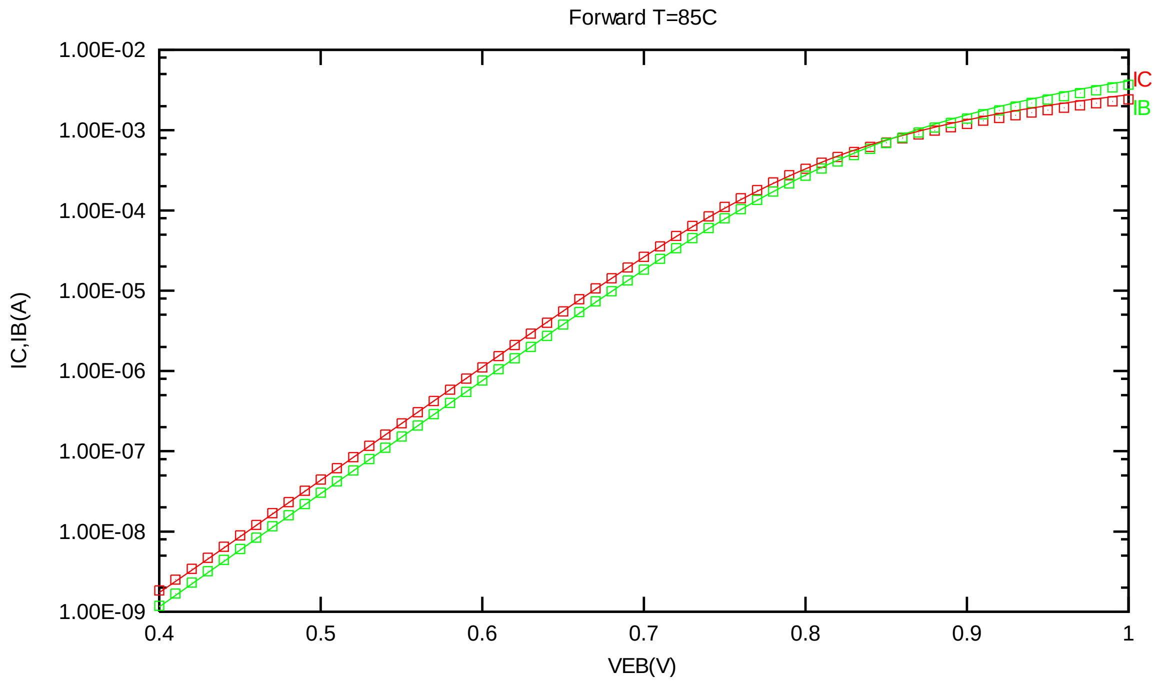

Figure 2-39 Forward gummel plot of PNP_SV45X45_L110E at 85 °C, VCB = -3.3V

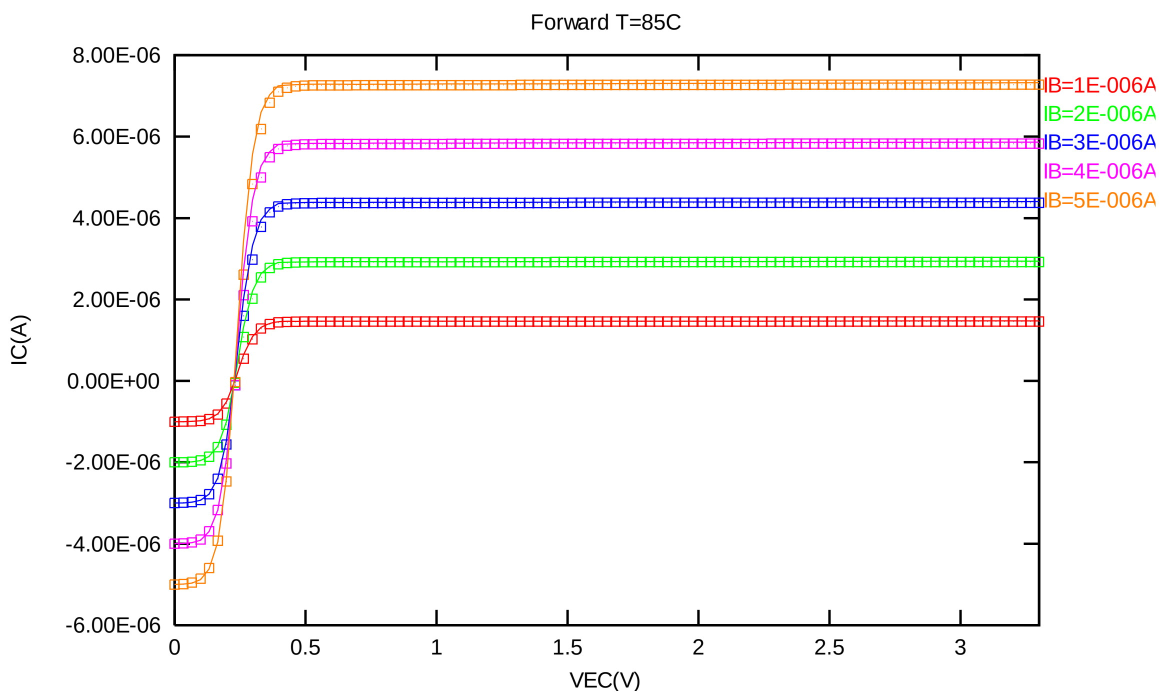

Figure 2-40 Forward output characteristics of PNP_SV45X45_L110E at 85 °C

2.6.3 PNP_SV45X45_L110E model verification at 125 °C

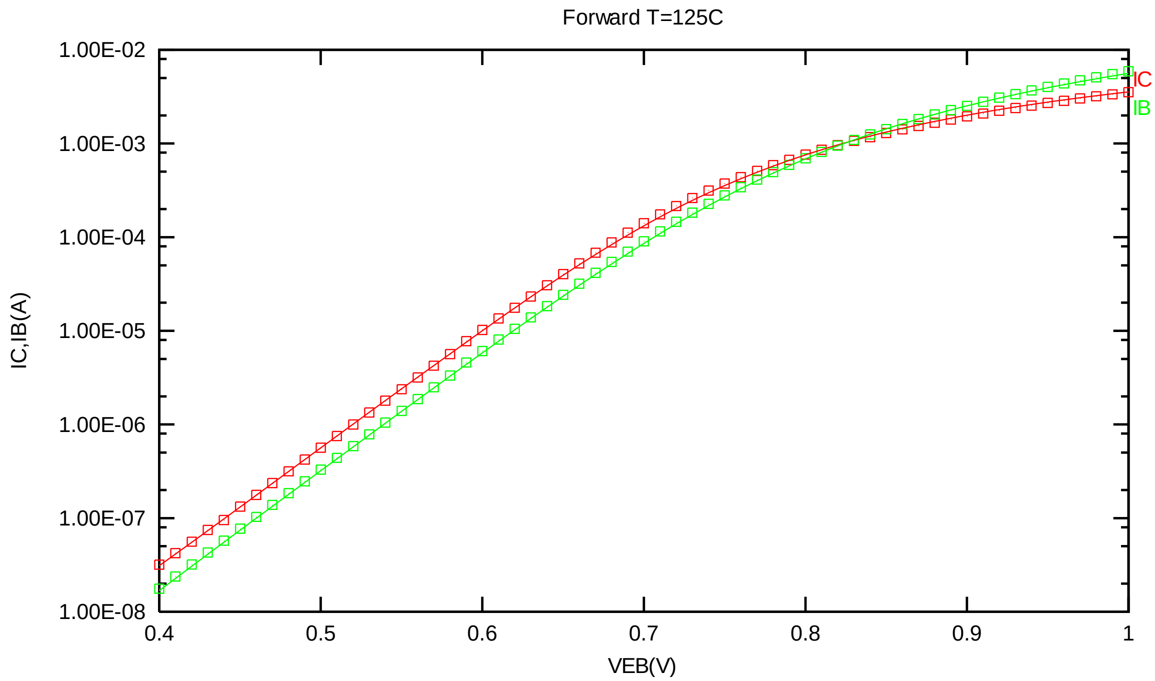

Figure 2-41 Forward gummel plot of PNP_SV45X45_L110E at 125 °C, VCB = -3.3V

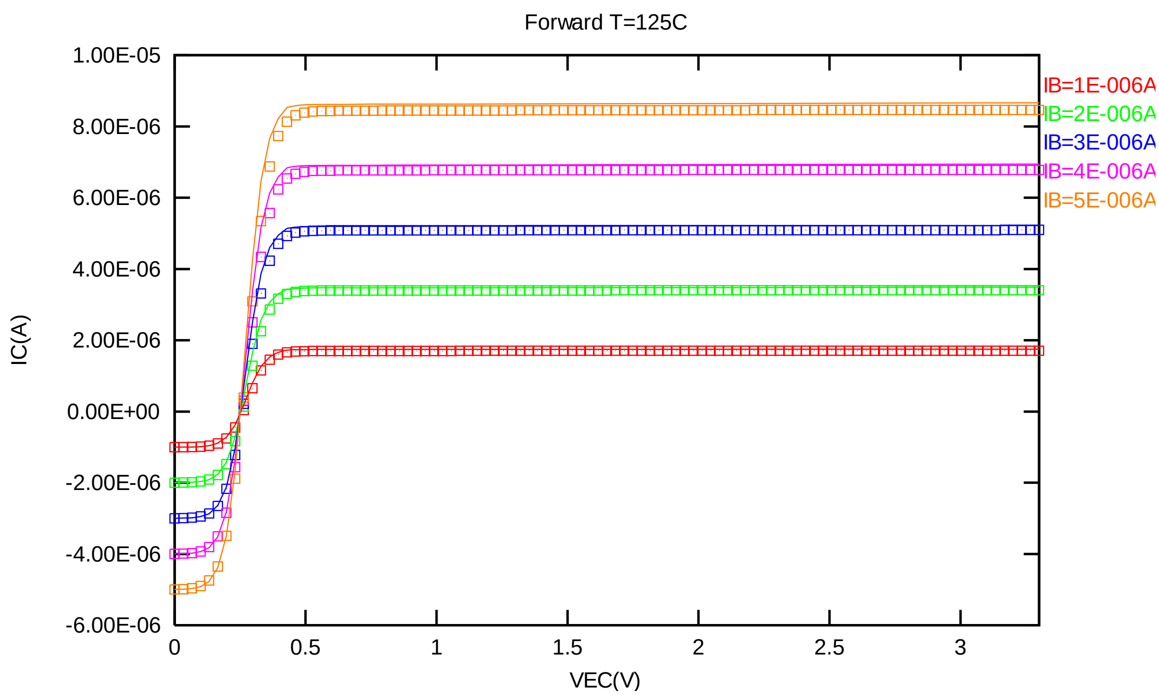

Figure 2-42 Forward output characteristics of PNP_SV45X45_L110E at 125 °C

2.6.4 PNP_SV45X45_L110E model verification at 0 °C

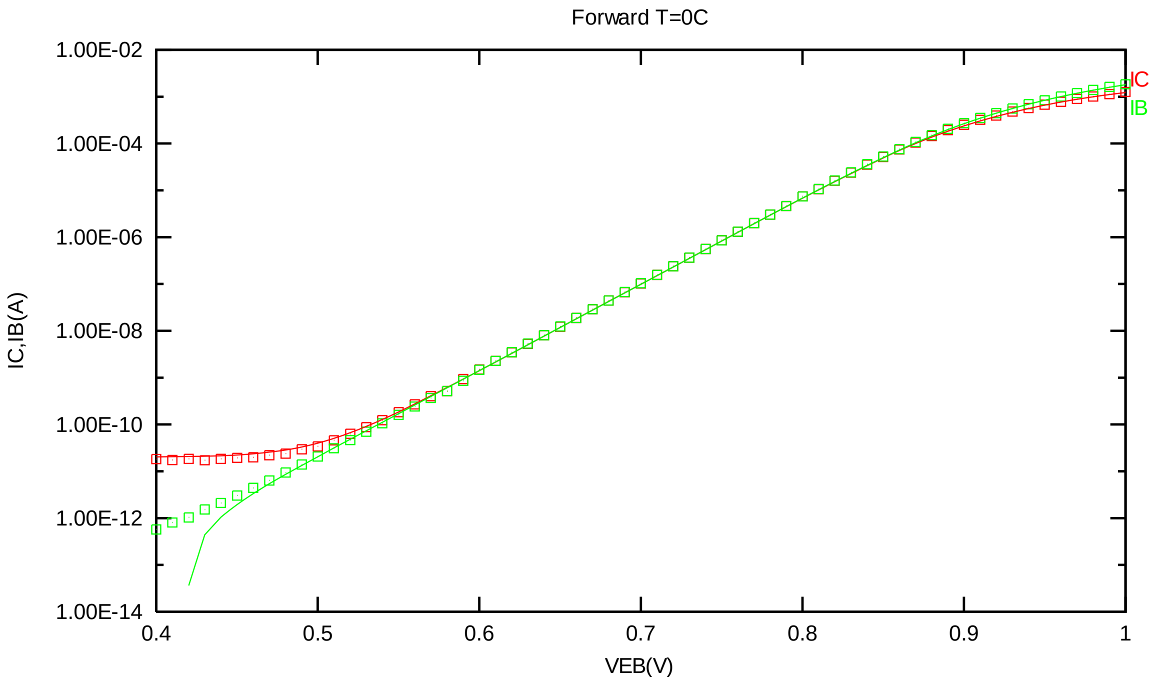

Figure 2-43 Forward gummel plot of PNP_SV45X45_L110E at 0 °C, VCB = -3.3V

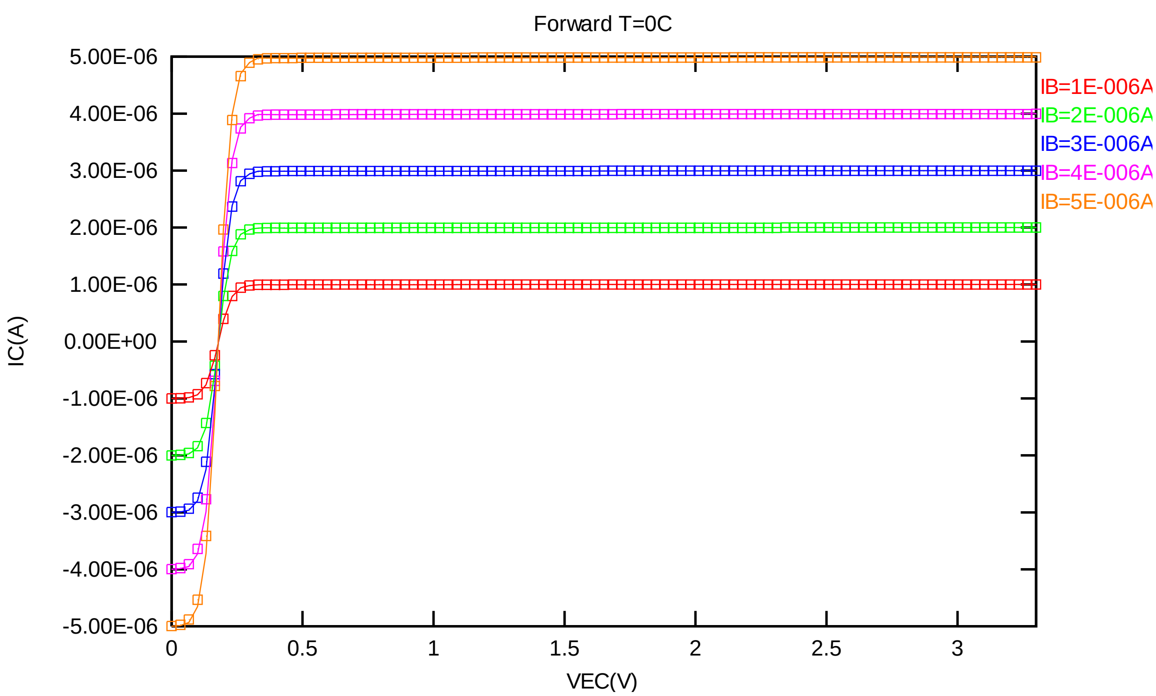

Figure 2-44 Forward output characteristics of PNP_SV45X45_L110E at 0 °C

2.6.5 PNP_SV45X45_L110E model verification at -50 °C

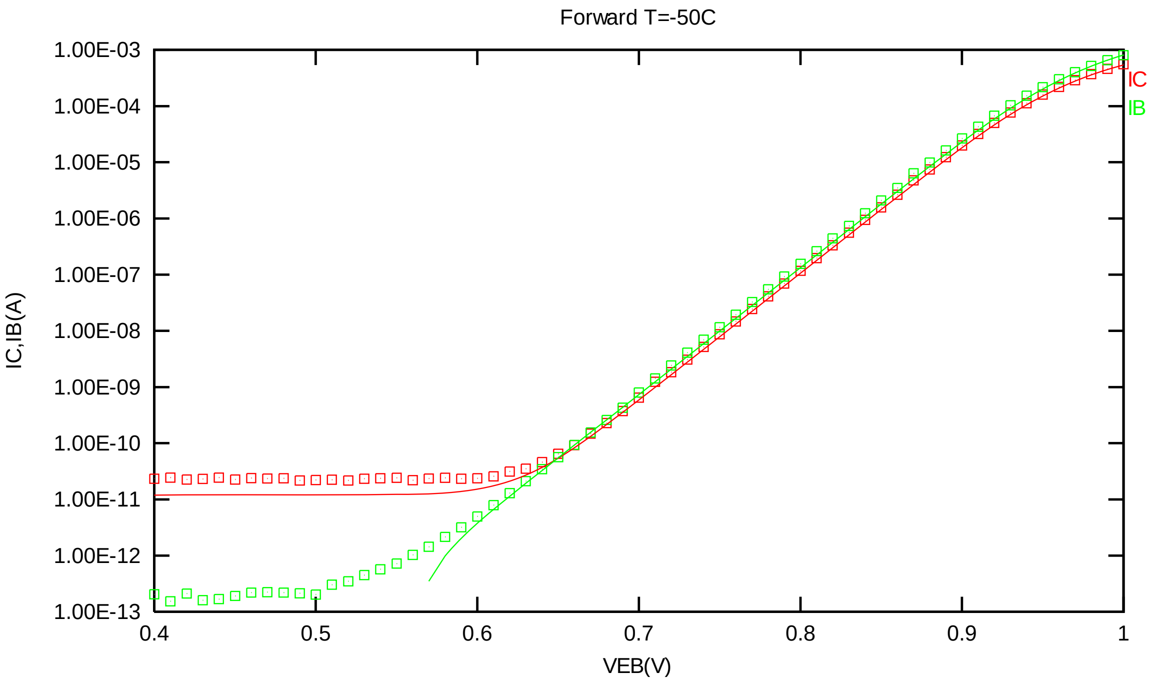

Figure 2-45 Forward gummel plot of PNP_SV45X45_L110E at -50 °C, VCB = -3.3V

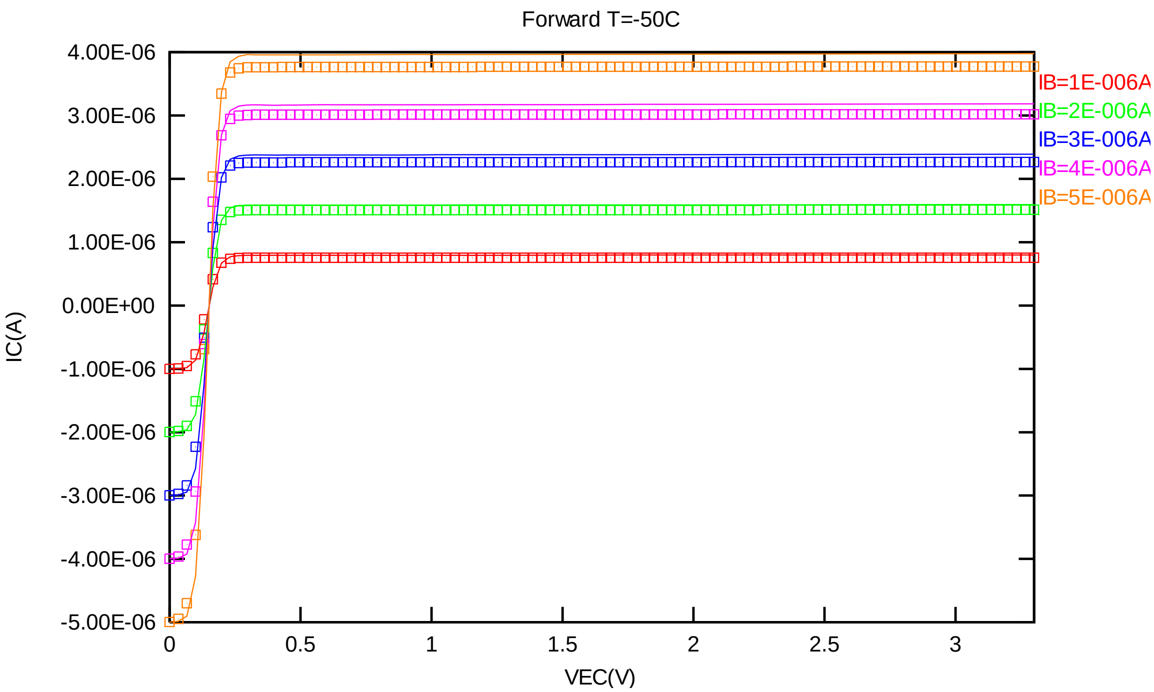

Figure 2-46 Forward output characteristics of PNP_SV45X45_L110E at -50 °C

2.6.6 PNP_SV45X45_L110E PNP model temperature effect

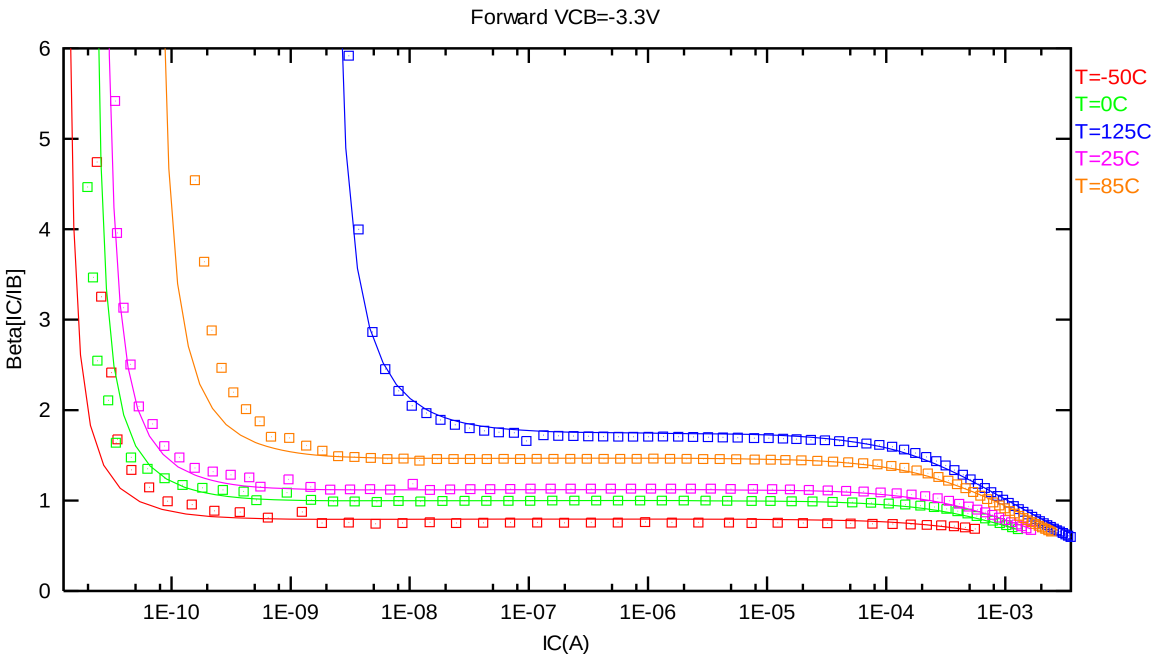

Figure 2-47 Current gain vs. IC at different temperatures of PNP_SV45X45_L110E, VCB = -3.3V

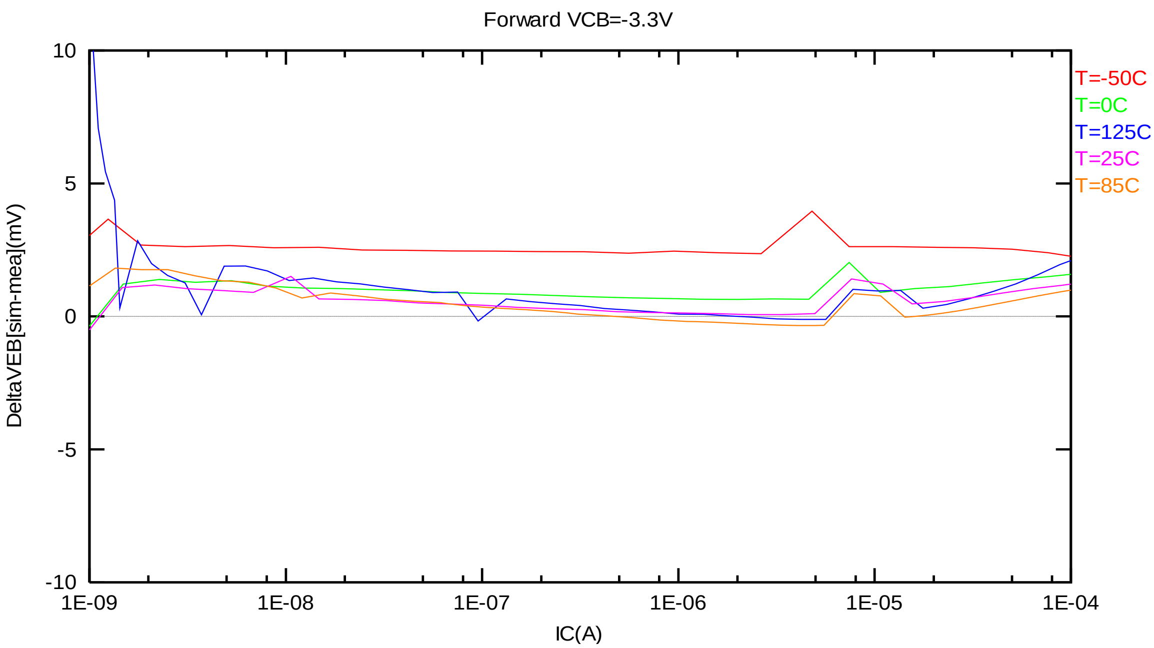

Figure 2-48 DeltaVEB vs. IC at different temperatures of PNP_SV45X45_L110E, VCB = -3.3V

2.7 PNP_SV90X90_L110E Model Verification

2.7.1 PNP_SV90X90_L110E model verification at 25 °C

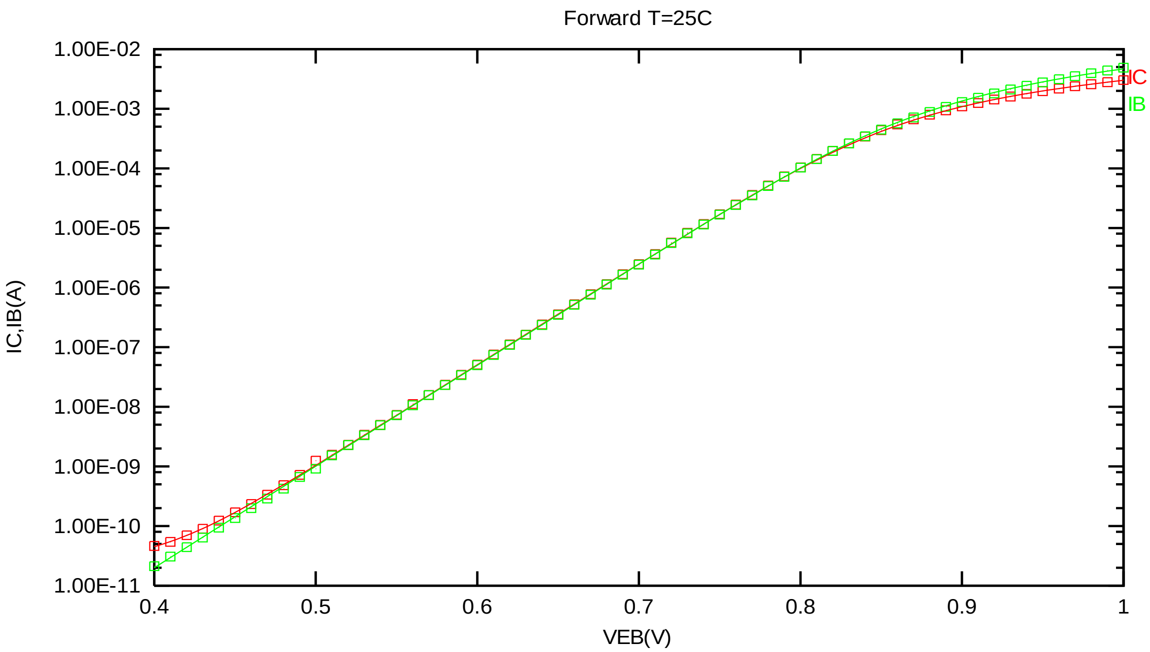

Figure 2-49 Forward gummel plot of PNP_SV90X90_L110E at 25 °C, VCB = -3.3V

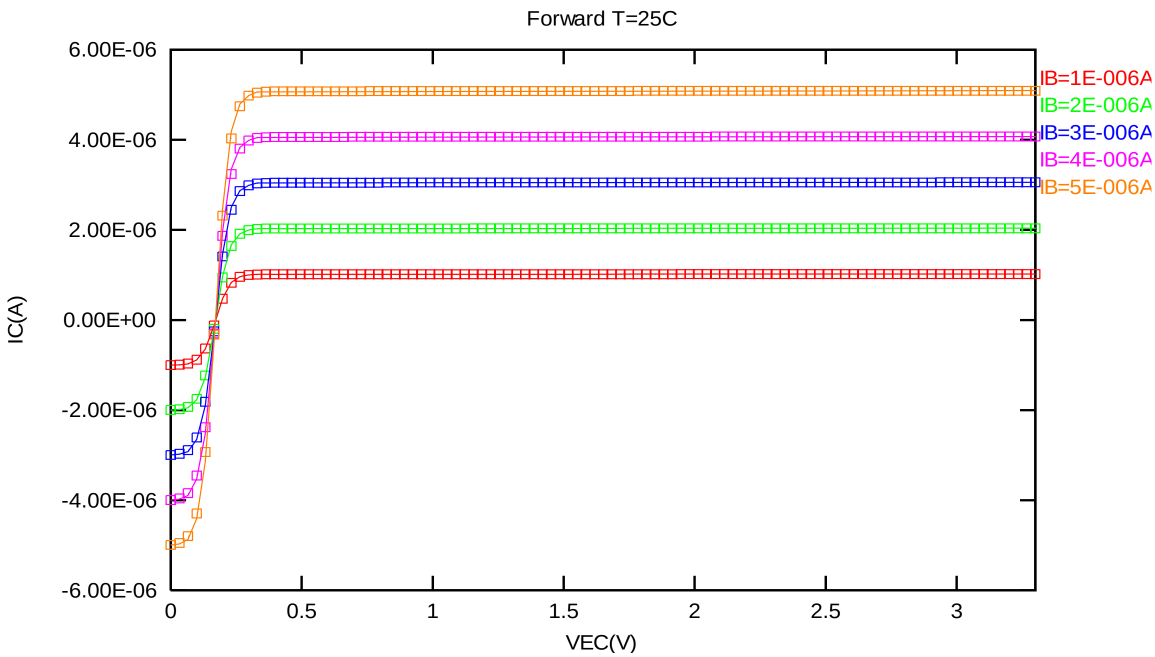

Figure 2-50 Forward output characteristics of PNP_SV90X90_L110E at 25 °C

2.7.2 PNP_SV90X90_L110E model verification at 85 °C

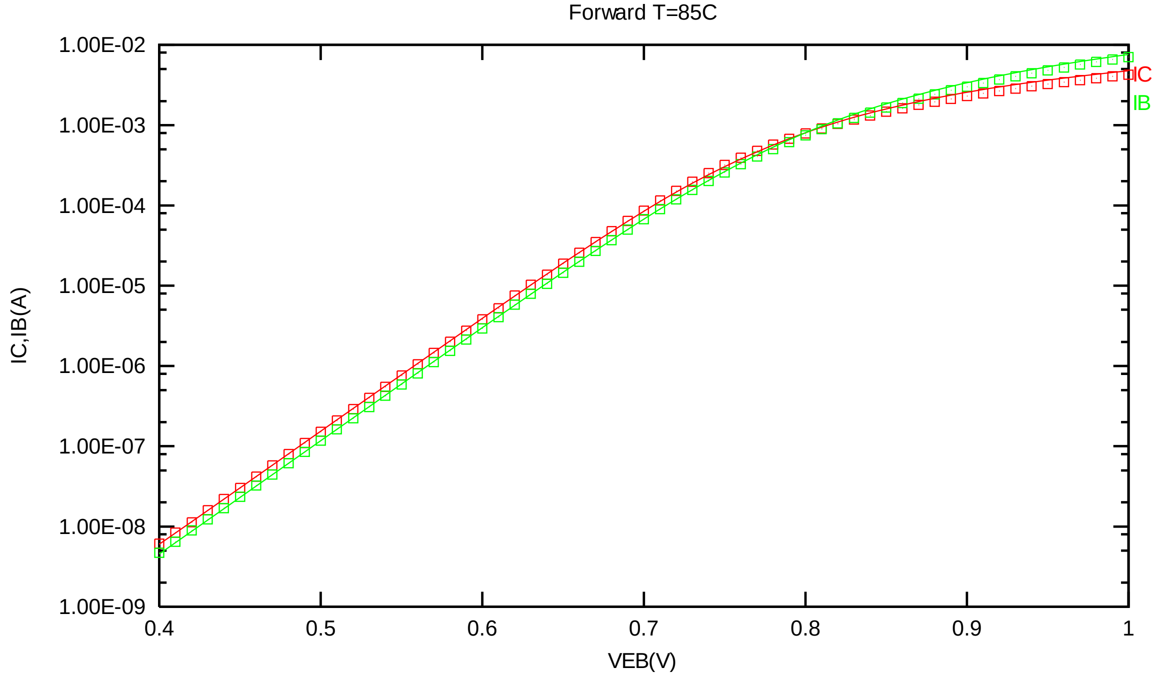

Figure 2-51 Forward gummel plot of PNP_SV90X90_L110E at 85 °C, VCB = -3.3V

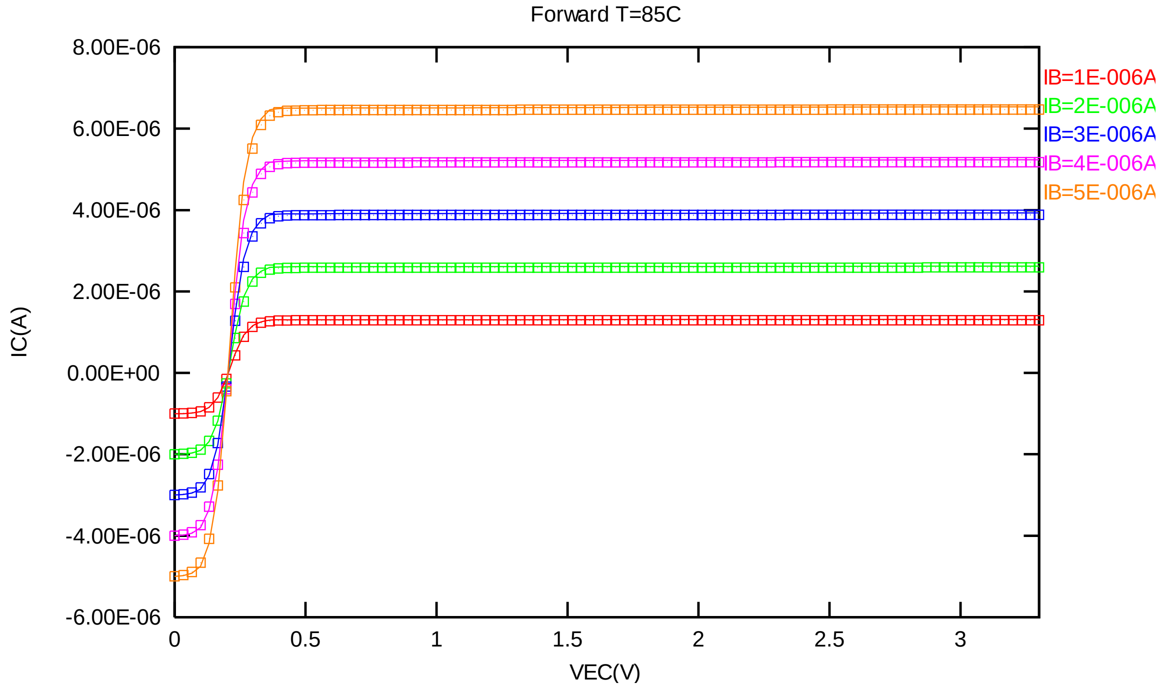

Figure 2-52 Forward output characteristics of PNP_SV90X90_L110E at 85 °C

2.7.3 PNP_SV90X90_L110E model verification at 125 °C

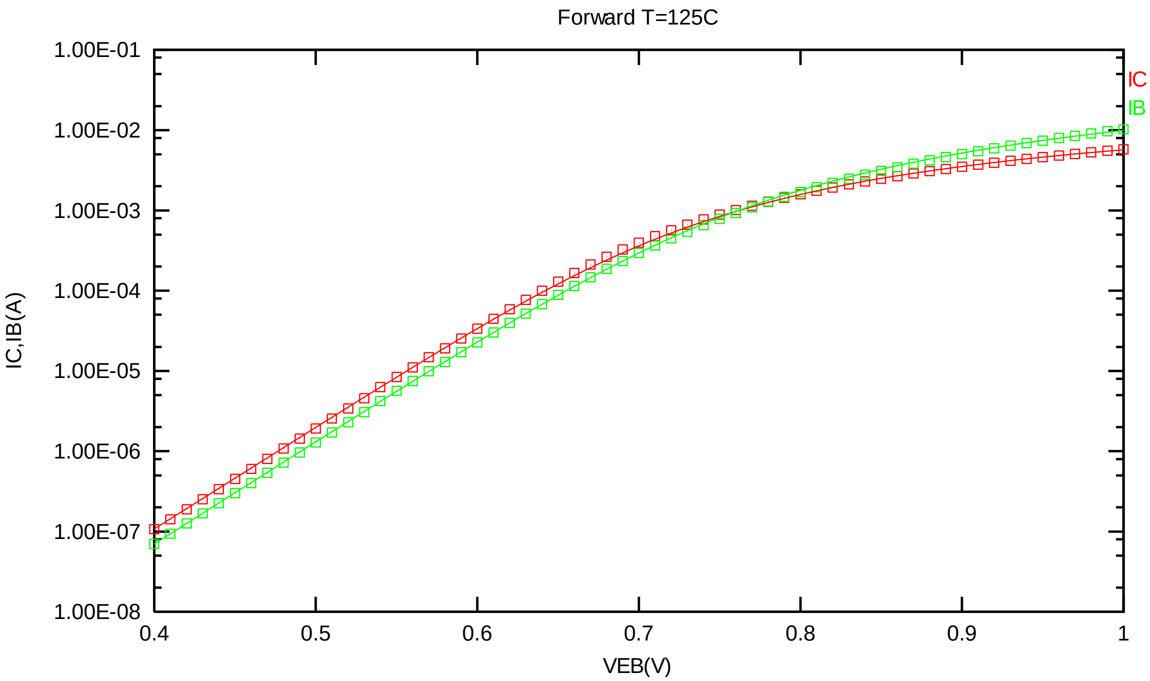

Figure 2-53 Forward gummel plot of PNP_SV90X90_L110E at 125 °C, VCB = -3.3V

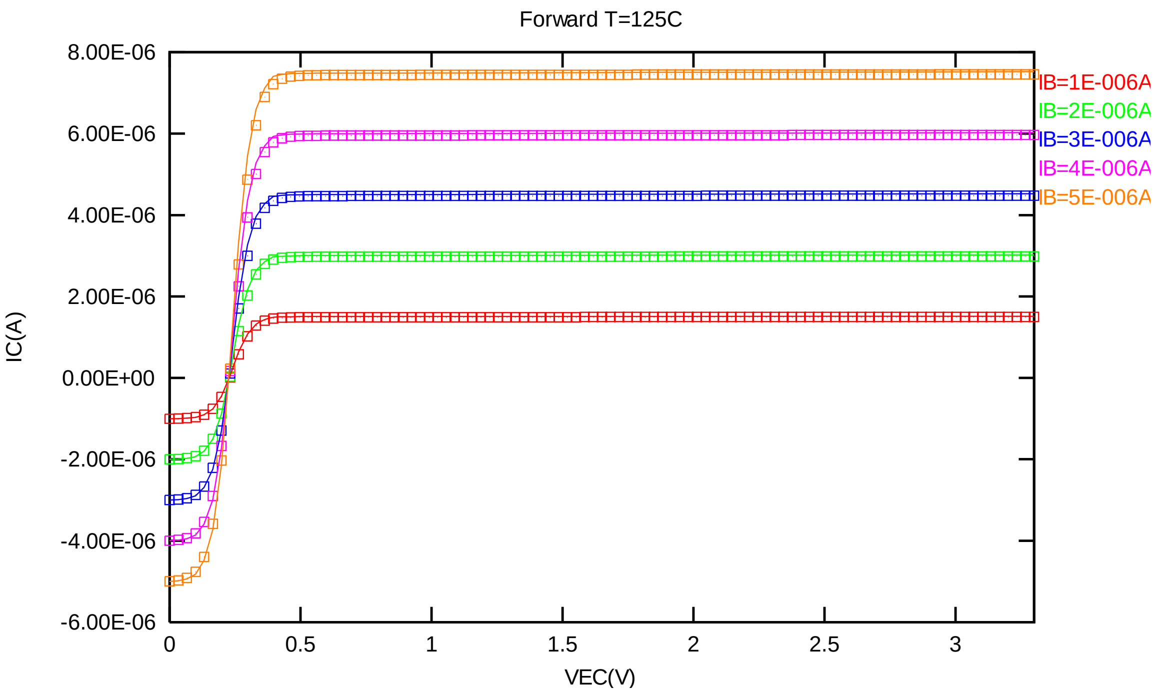

Figure 2-54 Forward output characteristics of PNP_SV90X90_L110E at 125 °C

2.7.4 PNP_SV90X90_L110E model verification at 0 °C

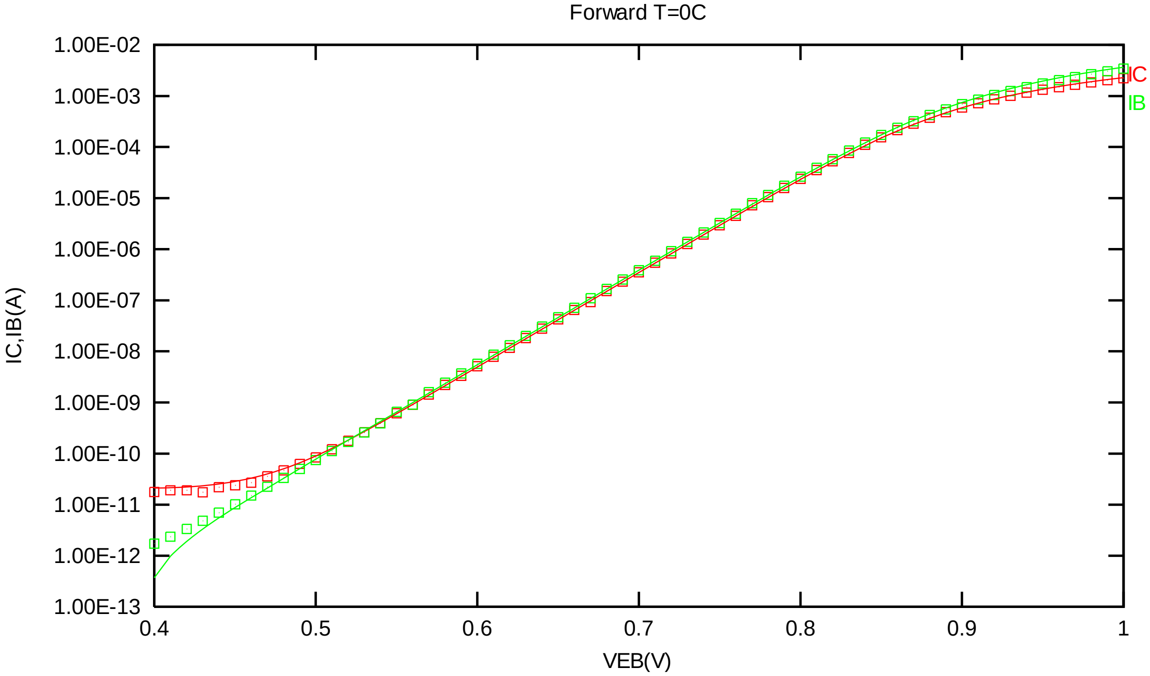

Figure 2-55 Forward gummel plot of PNP_SV90X90_L110E at 0 °C, VCB = -3.3V

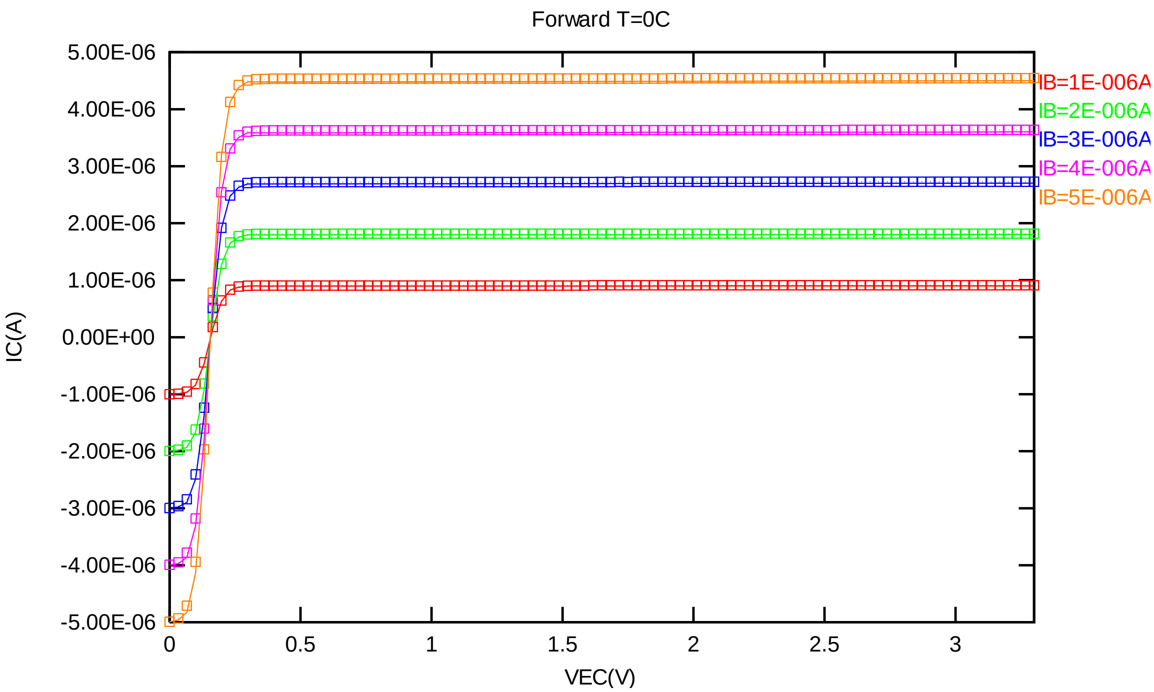

Figure 2-56 Forward output characteristics of PNP_SV90X90_L110E at 0 °C

2.7.5 PNP_SV90X90_L110E model verification at -50 °C

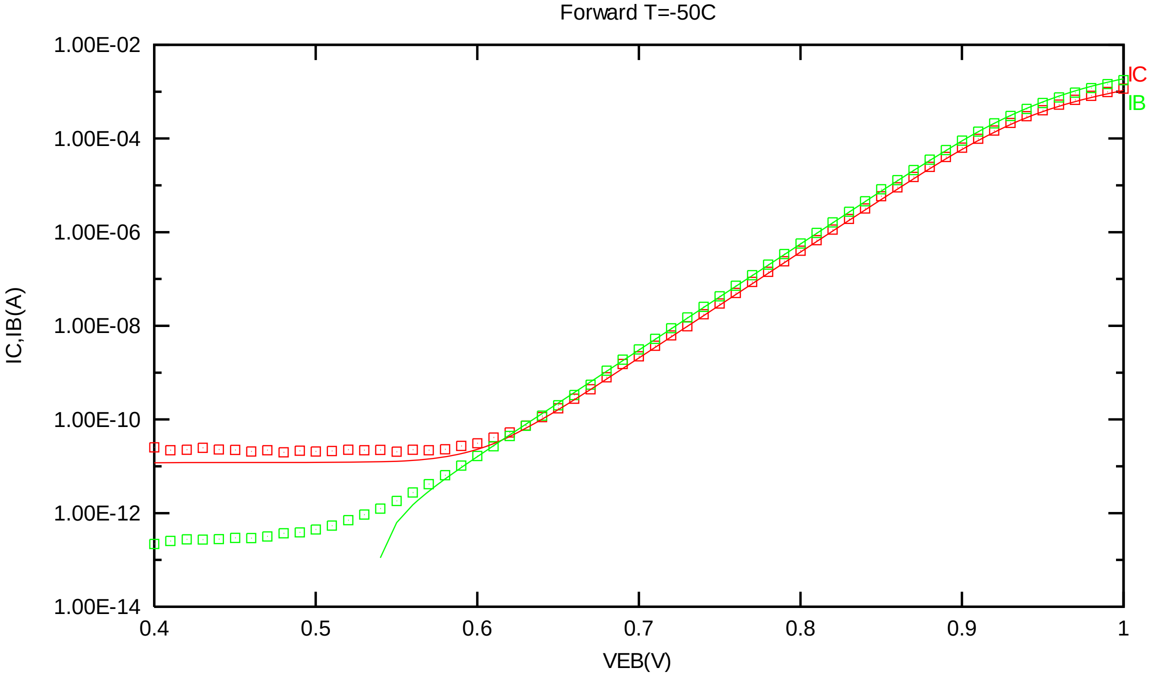

Figure 2-57 Forward gummel plot of PNP_SV90X90_L110E at -50 °C, VCB = -3.3V

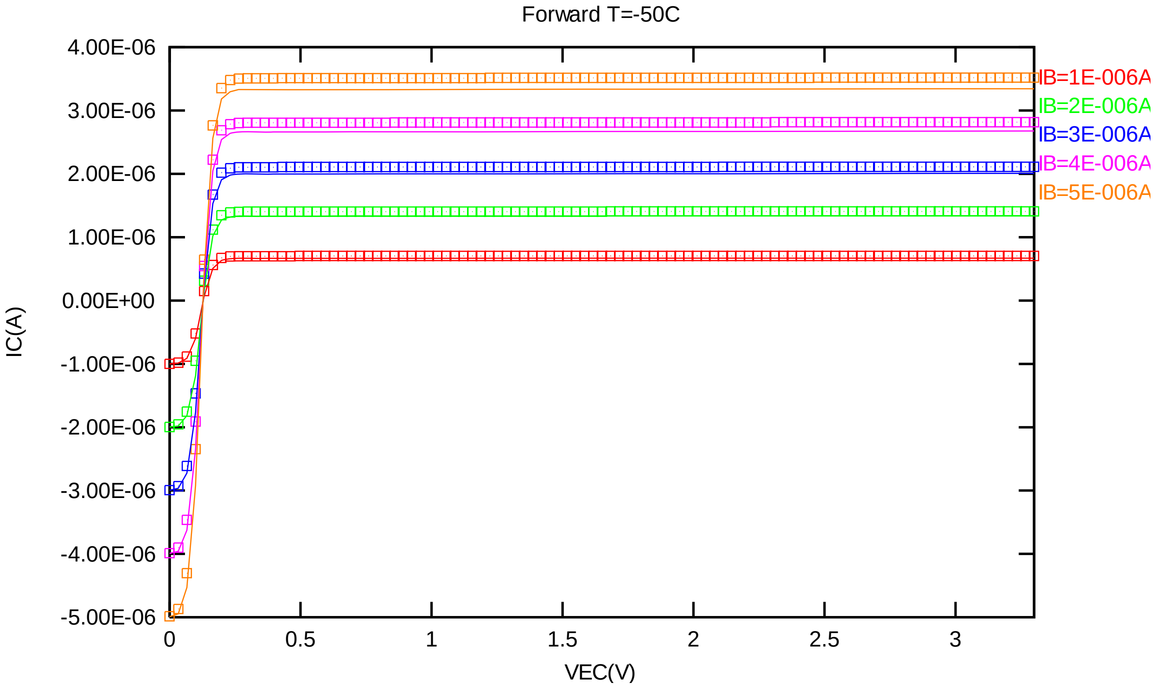

Figure 2-58 Forward output characteristics of PNP_SV90X90_L110E at -50 °C

2.7.6 PNP_SV90X90_L110E PNP model temperature effect

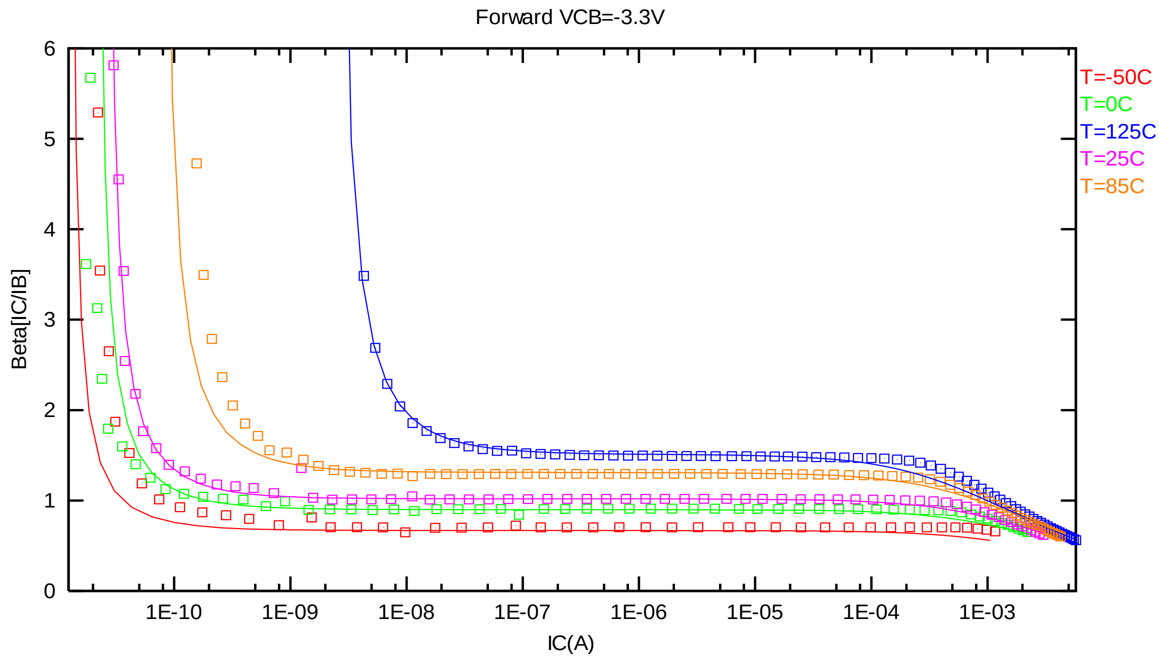

Figure 2-59 Current gain vs. IC at different temperatures of PNP_SV90X90_L110E, VCB = -3.3V

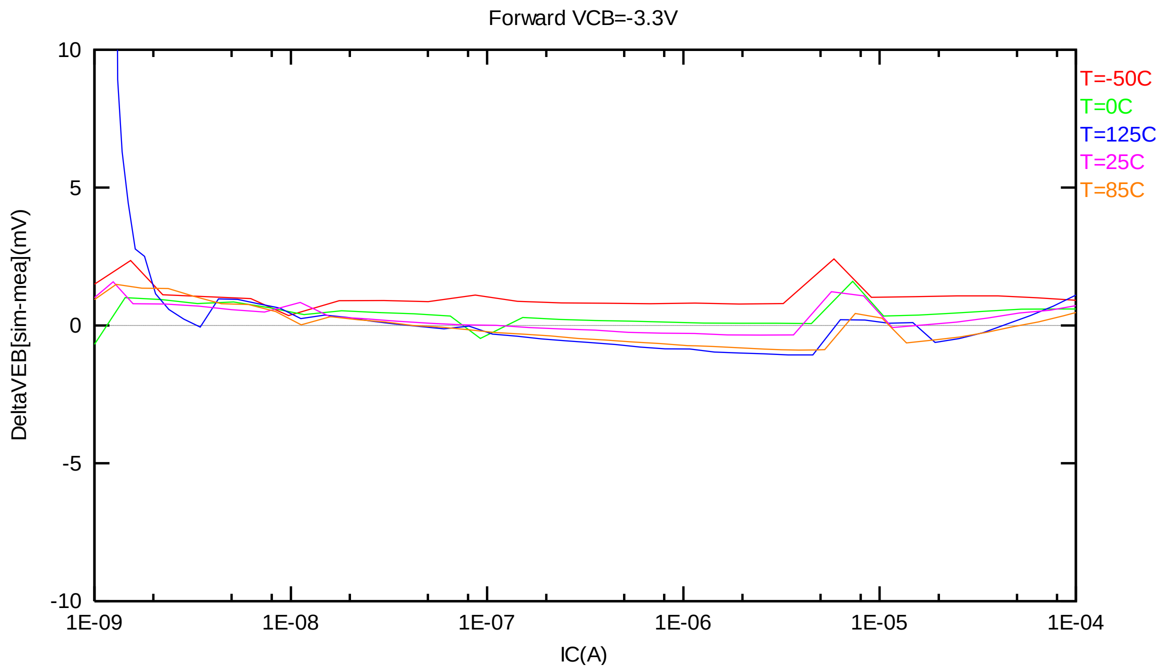

Figure 2-60 DeltaVEB vs. IC at different temperatures of PNP_SV90X90_L110E, VCB = -3.3V

2.8 PNP_SV135X135_L110E Model Verification

2.8.1 PNP_SV135X135_L110E model verification at 25 °C

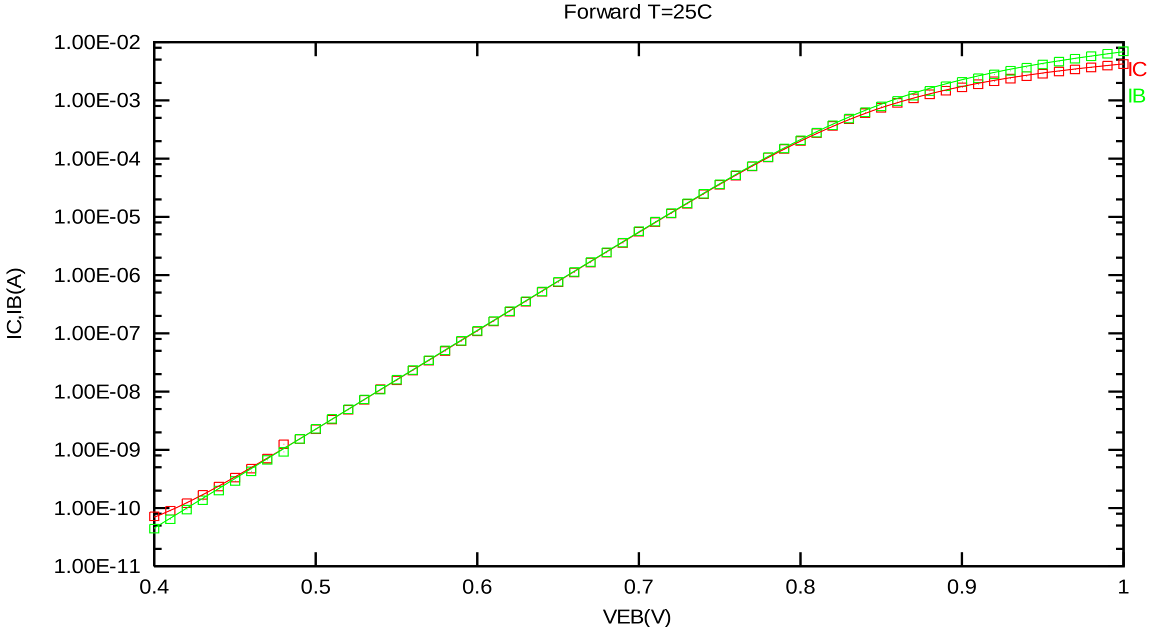

Figure 2-61 Forward gummel plot of PNP_SV135X135_L110E at 25 °C, VCB = -3.3V

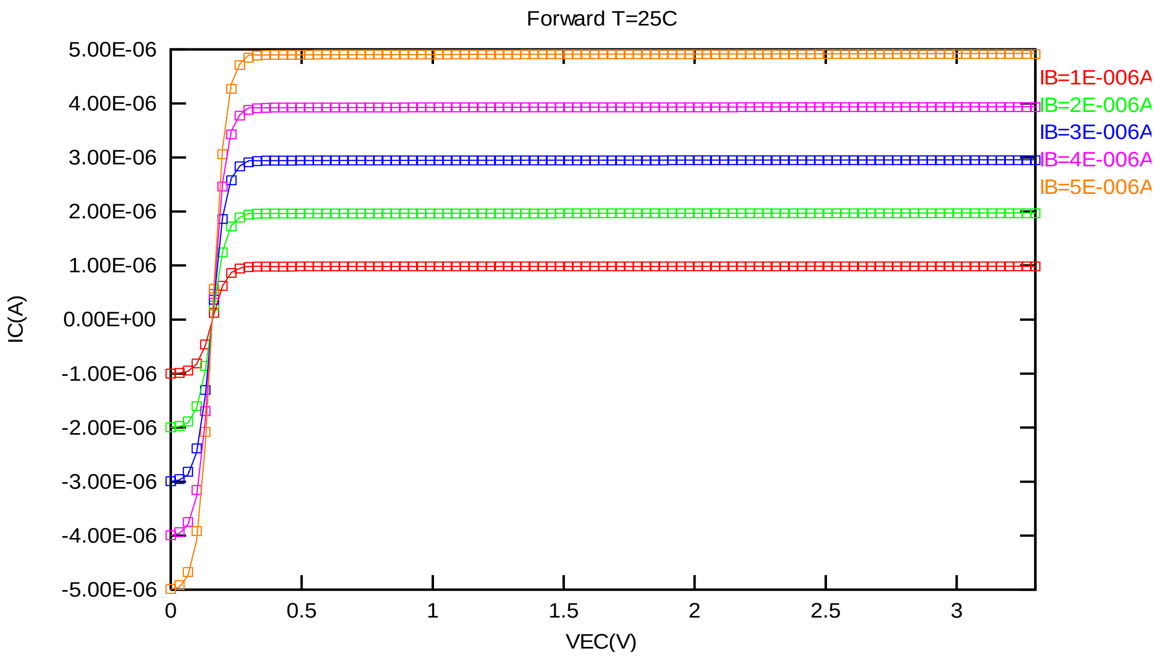

Figure 2-62 Forward output characteristics of PNP_SV135X135_L110E at 25 °C

2.8.2 PNP_SV135X135_L110E model verification at 85 °C

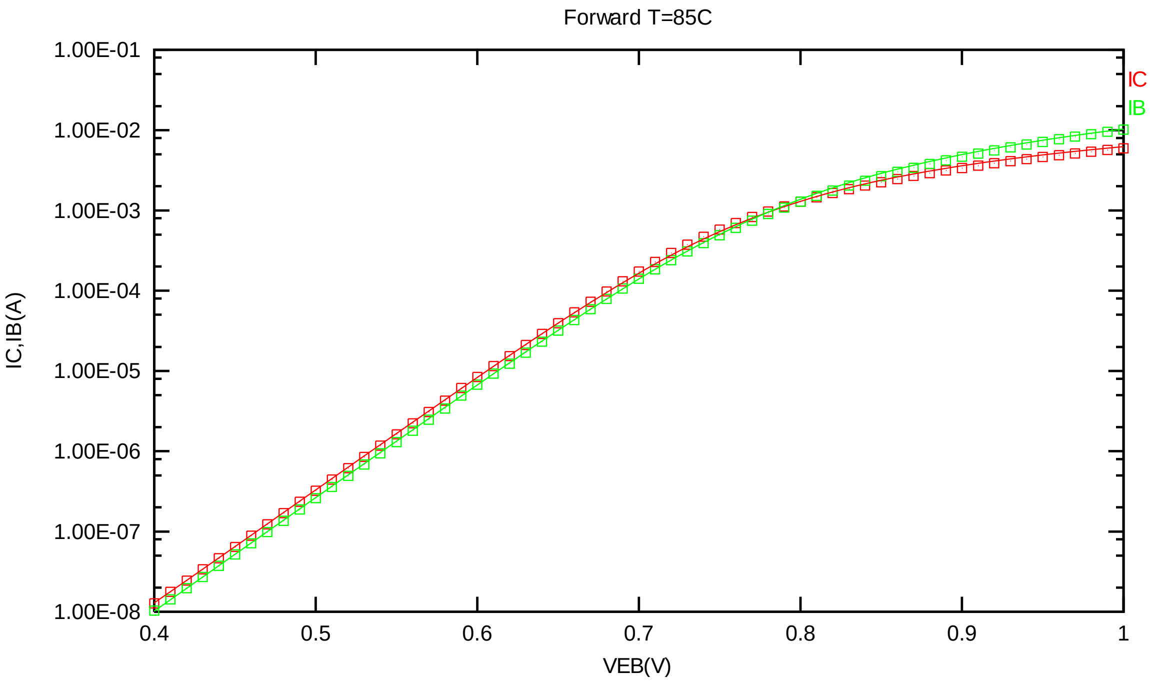

Figure 2-63 Forward gummel plot of PNP_SV135X135_L110E at 85 °C, VCB = -3.3V

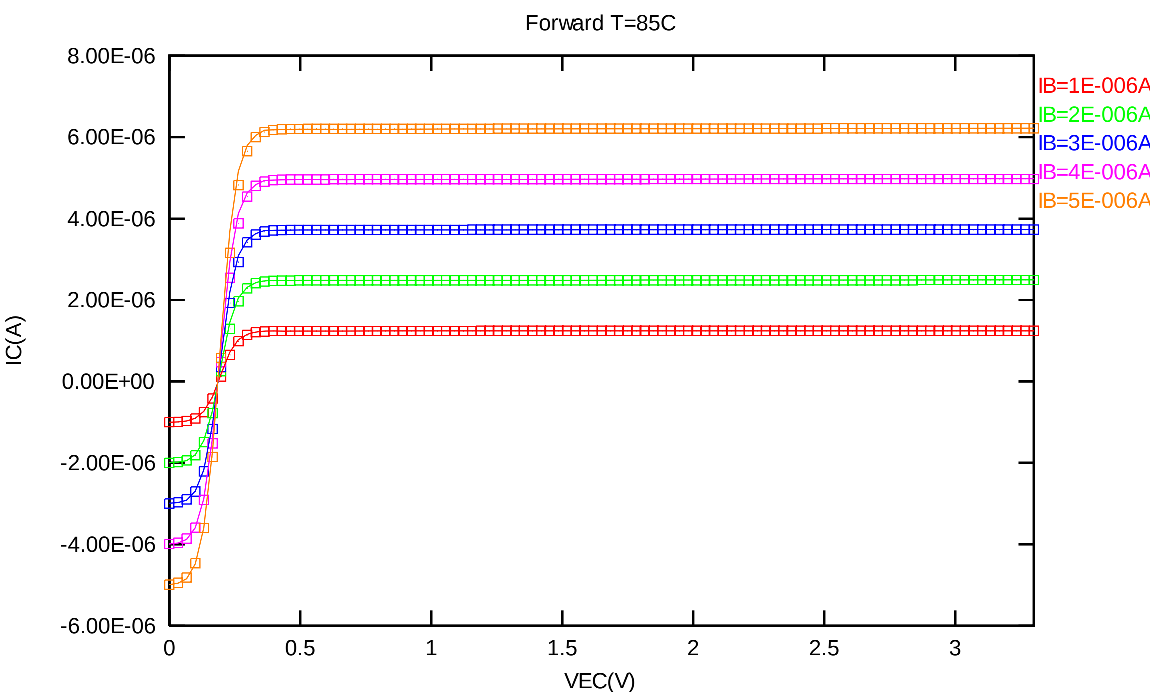

Figure 2-64 Forward output characteristics of PNP_SV135X135_L110E at 85 °C

2.8.3 PNP_SV135X135_L110E model verification at 125 °C

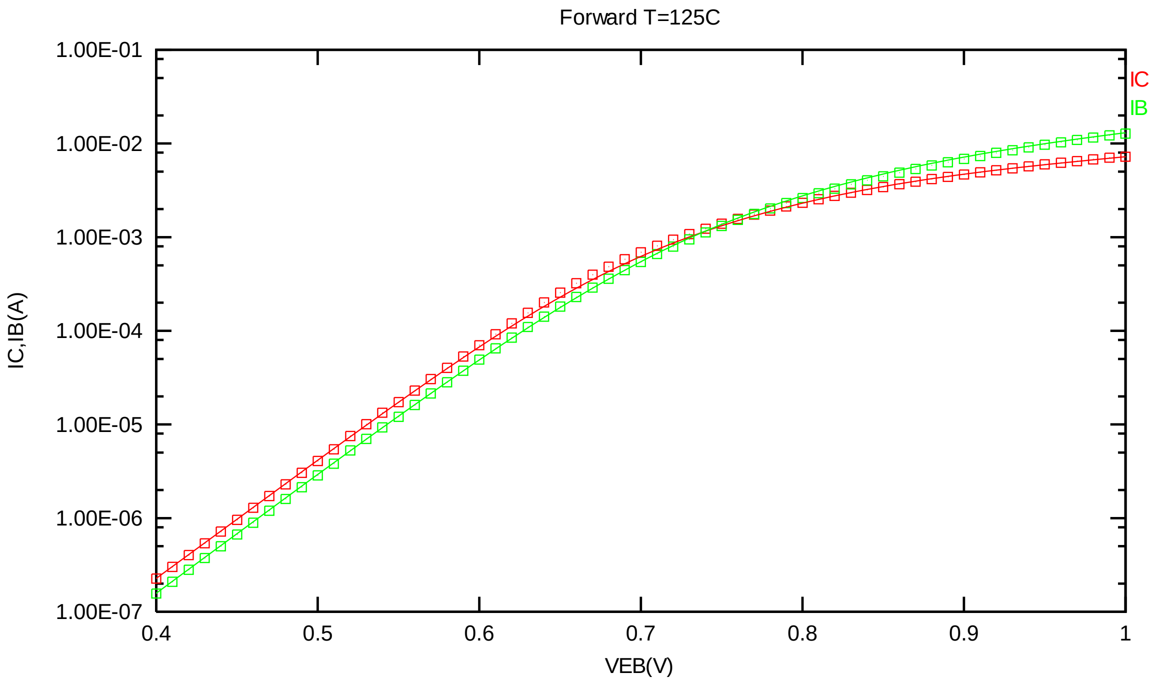

Figure 2-65 Forward gummel plot of PNP_SV135X135_L110E at 125 °C, VCB = -3.3V

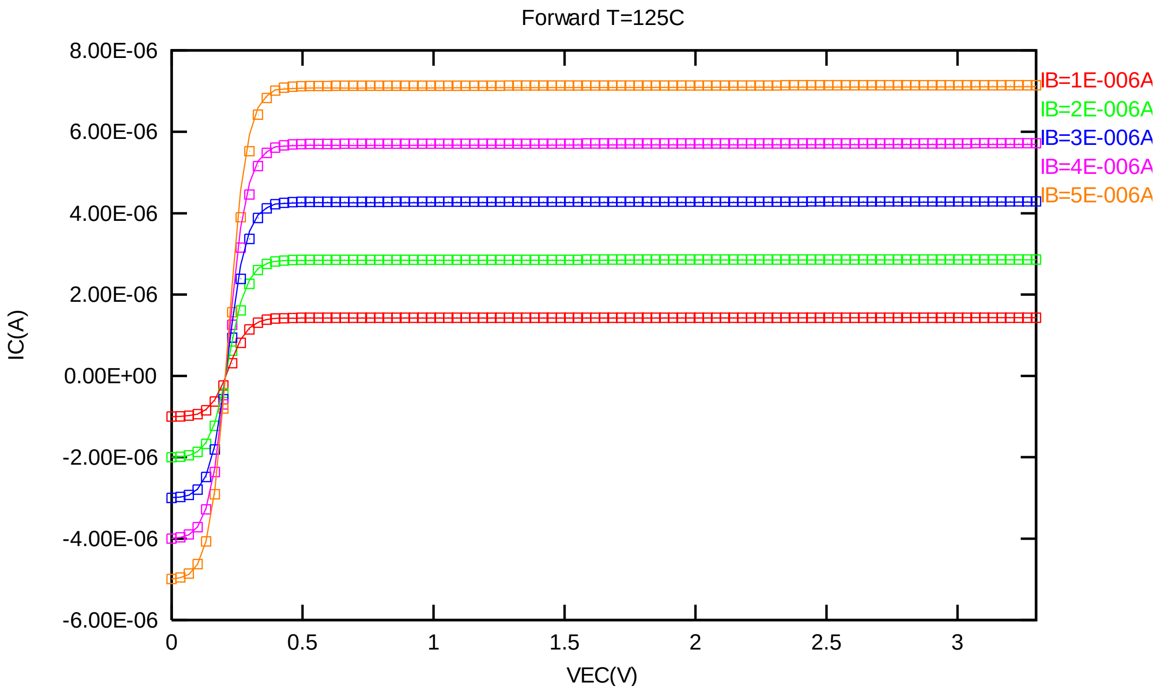

Figure 2-66 Forward output characteristics of PNP_SV135X135_L110E at 125 °C

2.8.4 PNP_SV135X135_L110E model verification at 0 °C

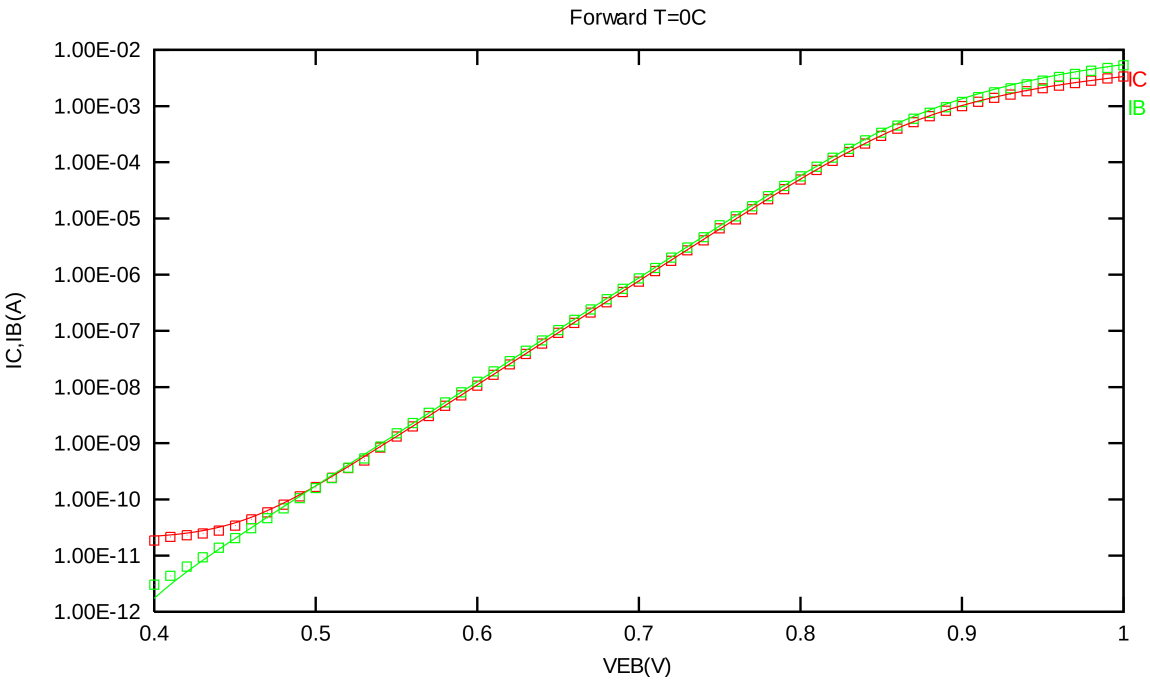

Figure 2-67 Forward gummel plot of PNP_SV135X135_L110E at 0 °C, VCB = -3.3V

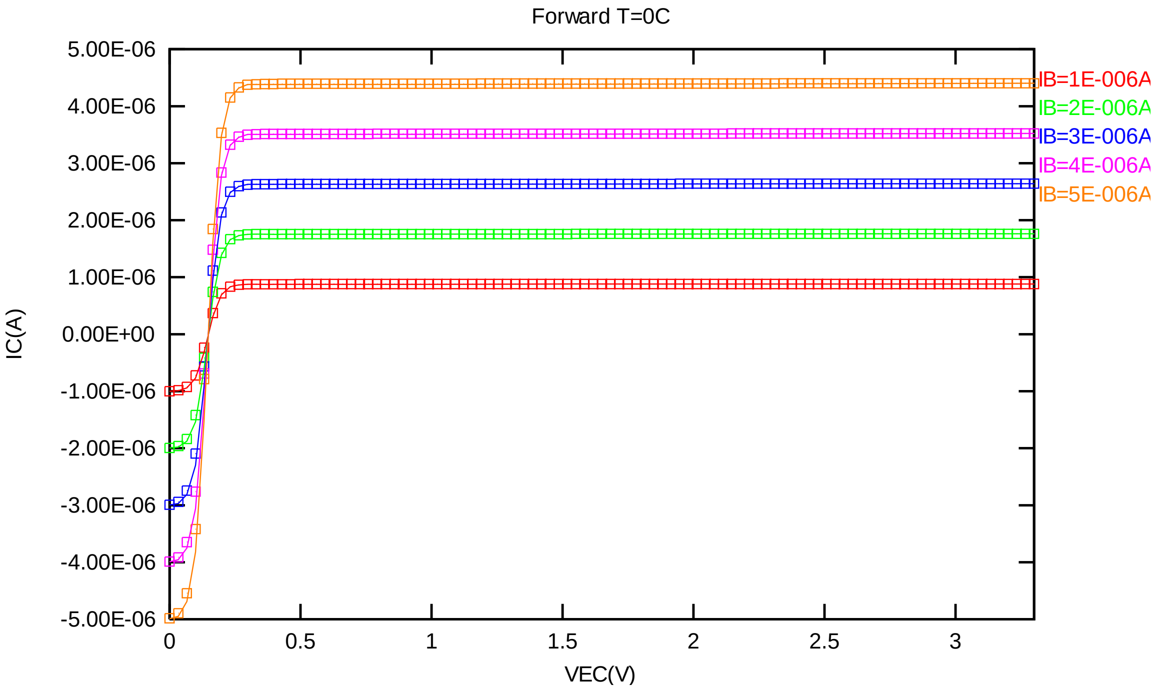

Figure 2-68 Forward output characteristics of PNP_SV135X135_L110E at 0 °C

2.8.5 PNP_SV135X135_L110E model verification at -50 °C

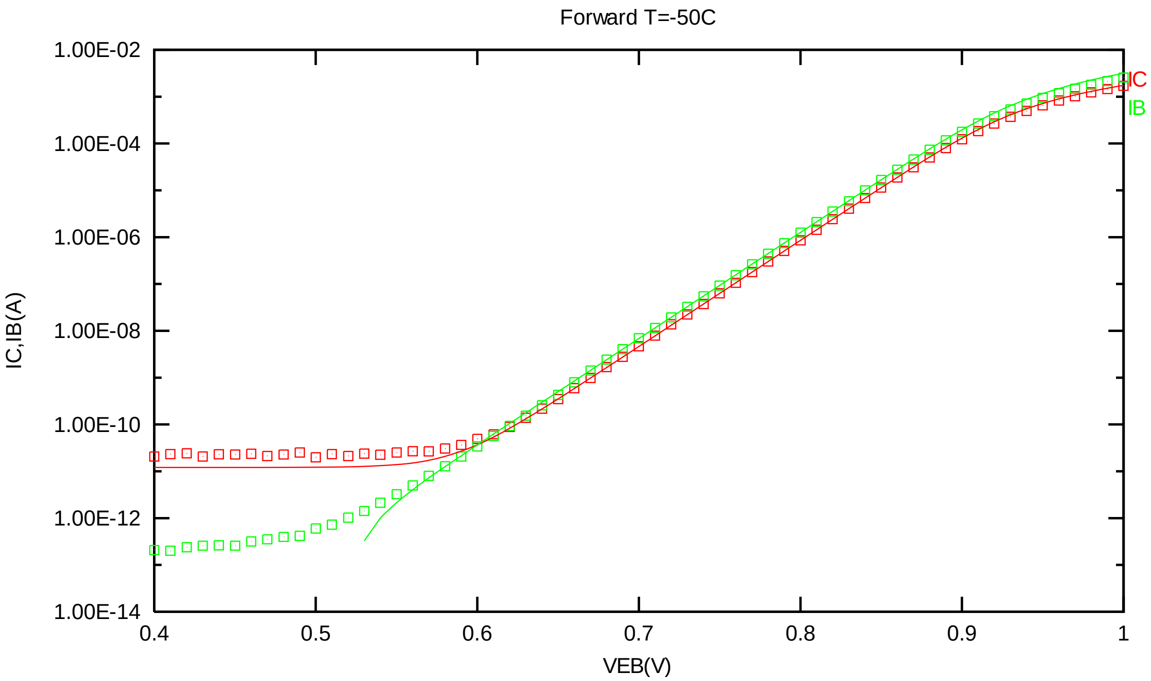

Figure 2-69 Forward gummel plot of PNP_SV135X135_L110E at -50 °C, VCB = -3.3V

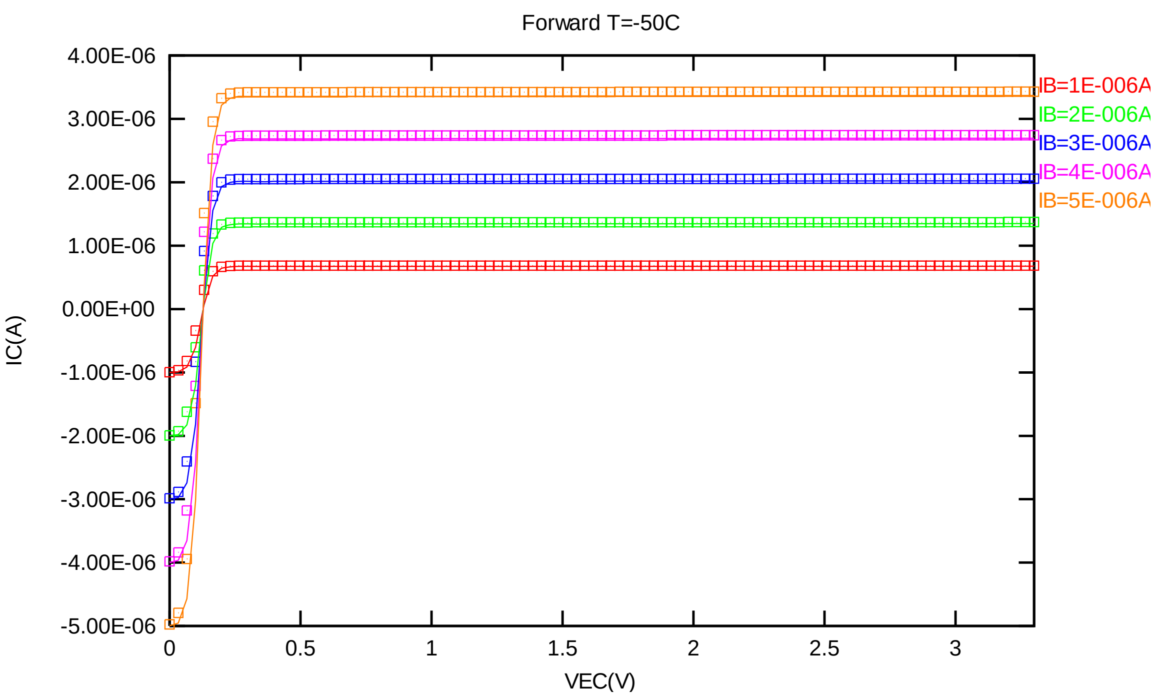

Figure 2-70 Forward output characteristics of PNP_SV135X135_L110E at -50 °C

2.8.6 PNP_SV135X135_L110E PNP model temperature effect

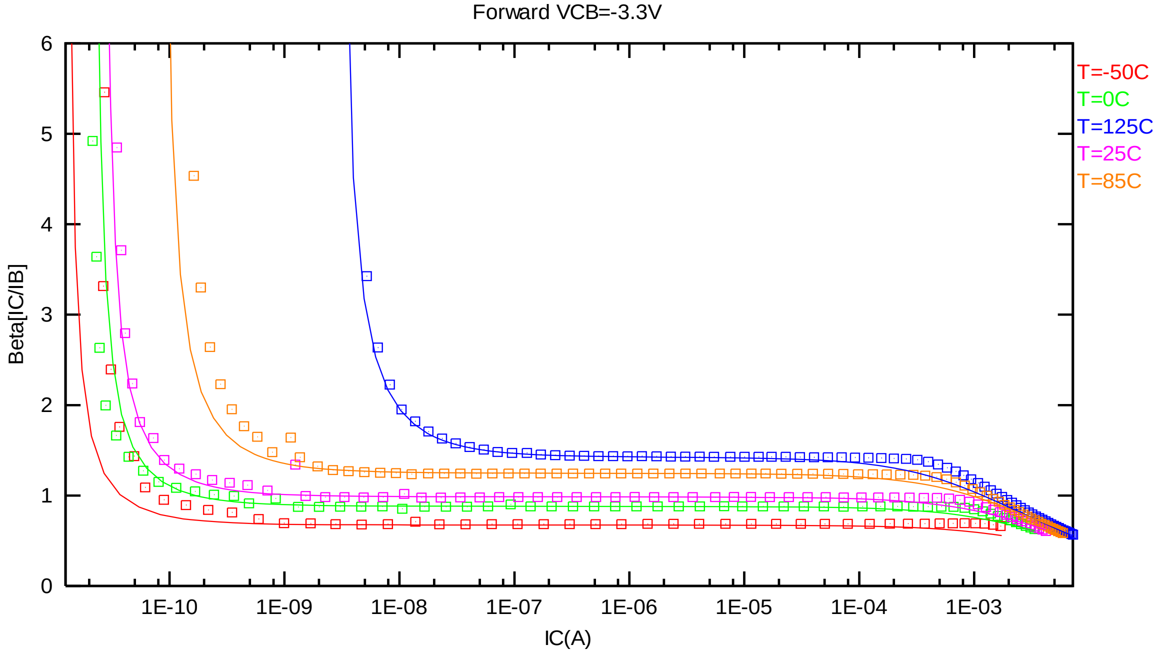

Figure 2-71 Current gain vs. IC at different temperatures of PNP_SV135X135_L110E, VCB = -3.3V

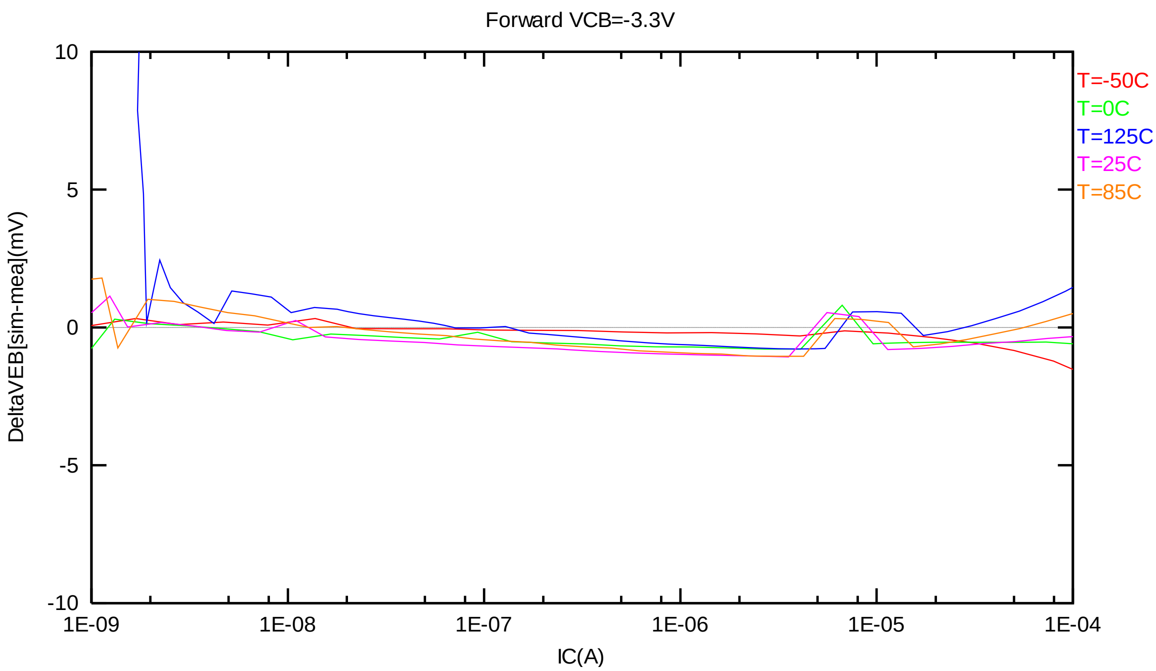

Figure 2-72 DeltaVEB vs. IC at different temperatures of PNP_SV135X135_L110E, VCB = -3.3V

3 Worst Case Model

3.1 Parameter Variations

Table 3-1 Corner parameters of PNP_V45X45_L110E

	Parameters	TT_BIP	FF_BIP	SS_BIP
1	IS	9.60E-19	X0.8000	X1.2000
2	RE	8.6181	X0.8000	X1.2000
3	RC	25.608	X0.8000	X1.2000
4	BF	1.121	X1.1500	X0.8500
5	NF	0.999	X0.9950	X1.0050
6	CJE	1.9622E-014	X0.9000	X1.1000
7	CJC	9.98E-15	X0.9000	X1.1000

Table 3-2 Corner parameters of PNP_V90X90_L110E

	Parameters	TT_BIP	FF_BIP	SS_BIP
1	IS	3.05E-18	X0.8000	X1.2000
2	RE	6.3241	X0.8000	X1.2000
3	RC	22.356	X0.8000	X1.2000
4	BF	1.0675	X1.1500	X0.8500
5	NF	0.997	X0.9950	X1.0050
6	CJE	7.8489E-014	X0.9000	X1.1000
7	CJC	1.98E-14	X0.9000	X1.1000

Table 3-3 Corner parameters of PNP_V135X135_L110E

	Parameters	TT_BIP	FF_BIP	SS_BIP
1	IS	6.82E-18	X0.8000	X1.2000
2	RE	2.5476	X0.8000	X1.2000
3	RC	22.815	X0.8000	X1.2000
4	BF	1.045	X1.1500	X0.8500
5	NF	0.999	X0.9950	X1.0050
6	CJE	1.766E-013	X0.9000	X1.1000
7	CJC	3.31E-14	X0.9000	X1.1000

Table 3-4 Corner parameters of PNP_SV45X45_L110E

	Parameters	TT_BIP	FF_BIP	SS_BIP
1	IS	9.5298E-019	X0.8000	X1.2000
2	RE	8.7616	X0.8000	X1.2000
3	RC	23.321	X0.8000	X1.2000
4	BF	1.1497	X1.1500	X0.8500
5	NF	0.99574	X0.9950	X1.0050
6	CJE	1.9622E-014	X0.9000	X1.1000
7	CJC	1.1334E-014	X0.9000	X1.1000

Table 3-5 Corner parameters of PNP_SV90X90_L110E

	Parameters	TT_BIP	FF_BIP	SS_BIP
1	IS	3.3706E-018	X0.8000	X1.2000
2	RE	4.8251	X0.8000	X1.2000
3	RC	22.231	X0.8000	X1.2000
4	BF	1.0405	X1.1500	X0.8500
5	NF	0.99574	X0.9950	X1.0050
6	CJE	7.8489E-014	X0.9000	X1.1000
7	CJC	2.173E-014	X0.9000	X1.1000

Table 3-6 Corner parameters of PNP_SV135X135_L110E

	Parameters	TT_BIP	FF_BIP	SS_BIP
1	IS	7.3999E-018	X0.8000	X1.2000
2	RE	2.9303	X0.8000	X1.2000
3	RC	22.114	X0.8000	X1.2000
4	BF	1.0053	X1.1500	X0.8500
5	NF	0.99574	X0.9950	X1.0050
6	CJE	1.766E-013	X0.9000	X1.1000
7	CJC	3.5479E-014	X0.9000	X1.1000

4 Appendix

4.1 HSPICE Warning Message

Table 4-1 HSPICE warning message

Item	Warning Message	Explanation
1.	None	None

4.2 ELDO Warning Message

Table 4-2 ELDO warning message

Item	Warning Message	Explanation
1.	None	None

4.3 SPECTRE Warning Message

Table 4-3 SPECTRE warning message

Item	Warning Message	Explanation
1.	None	None